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Chapter 1

Methods and Preliminaries

1.1 The Unitary renormalisation group method

The URG method was introduced and formalised in refs. [1–4]. This section is adapted from those references and expanded wherever required.

Description of the problem

We are given a Hamiltonian $\mathcal{H}$ which is not completely diagonal in the occupation number basis of the electrons, $\hat{n}_k$: $[\mathcal{H}, n_k] \neq 0$. k labels any set of quantum numbers depending on the system. For spin-less Fermions it can be the momentum of the particle, while for spin-full Fermions it can be the set of momentum and spin. There are terms that scatter electrons from one quantum number k to another quantum number k' .

We take a general Hamiltonian,

$$\mathcal{H} = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \quad (1.1)$$

Formally, we can decompose the entire Hamiltonian in the subspace of the electron we want to decouple ($q\beta$).

$$\mathcal{H} = \begin{pmatrix} |1\rangle & |0\rangle \\ H_1 & T \\ T^\dagger & H_0 \end{pmatrix} \quad (1.2)$$

The basis in which this matrix is written is $\{|1\rangle, |0\rangle\}$ where $|i\rangle$ is the set of all states where $\hat{n}_{q\beta} = i$. The aim of one step of the URG is to find a unitary transformation U such that the new Hamiltonian $U\mathcal{H}U^\dagger$ is diagonal in this already-chosen basis.

$$\tilde{\mathcal{H}} \equiv U\mathcal{H}U^\dagger = \begin{pmatrix} |1\rangle & |0\rangle \\ \tilde{H}_1 & 0 \\ 0 & \tilde{H}_0 \end{pmatrix} \quad (1.3)$$

U_q is defined by

$$\tilde{\mathcal{H}} = U_q \mathcal{H} U_q^\dagger \text{ such that } [\tilde{\mathcal{H}}, n_q] = 0 \quad (1.4)$$

It is clear that U is the diagonalizing matrix for $\mathcal{H}$. Hence we can frame this problem as an eigenvalue equation as well. Let $|\psi_1\rangle, |\psi_0\rangle$ be the basis in which the original Hamiltonian $\mathcal{H}$ has no off-diagonal terms corresponding to $q\beta$. Hence, we can write

$$\mathcal{H} |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle, i \in \{0, 1\} \quad (1.5)$$

Since $|\psi_i\rangle$ is the set of eigenstates of $\mathcal{H}$ and $|i\rangle$ is the set of eigenstates in which $U\mathcal{H}U^\dagger$ has no off-diagonal terms corresponding to $q\beta$, we can relate $|\psi_i\rangle$ and $|i\rangle$ by the same transformation : $|\psi_i\rangle = U^\dagger |i\rangle$. We can expand the state $|\psi_i\rangle$ in the subspace of $q\beta$:

$$|\psi_i\rangle = \sum_{j=0,1} |j\rangle \langle j | \psi_i \rangle \equiv |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle \quad (1.6)$$

where $|\phi_j^i\rangle = \langle j | \psi_i \rangle$. If we substitute the expansion 1.2 into the eigenvalue equation 1.5, we get

$$\left[H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle \quad (1.7)$$

The diagonal parts $H_e = \text{tr} [\mathcal{H} \hat{n}_{q\beta}]$ and $H_h = \text{tr} [\mathcal{H} (1 - \hat{n}_{q\beta})]$ can be separated into a purely diagonal part $\mathcal{H}^d$ that contains the single-particle energies and the multi-particle correlation energies or Hartree-like contributions, and an off-diagonal part $\mathcal{H}^i$ that scatters between the remaining degrees of freedom $k\sigma \neq q\beta$. That is,

$$H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i$$

This gives

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\tilde{H}_i - \mathcal{H}^i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.8)$$

Obtaining the decoupling transformation

We now define a new operator $\hat{\omega}_i = \tilde{H}_i - \mathcal{H}^i$, such that

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\hat{\omega}_i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.9)$$

From the definition of $\hat{\omega}_i$, we can see that it is Hermitian and has no term that scatters in the subspace of $q\beta$, so it is diagonal in $q\beta$ and we can expand it as $\hat{\omega}_i = \hat{\omega}_i^1 \hat{n}_{q\beta} + \hat{\omega}_i^0 (1 - \hat{n}_{q\beta})$. Using the expansion 1.6, we can write

$$\hat{\omega}_i |\psi_i\rangle = \hat{\omega}_i^1 |1\rangle |\phi_1^i\rangle + \hat{\omega}_i^0 |0\rangle |\phi_0^i\rangle \quad (1.10)$$

Since the only requirement on $|\psi_i\rangle$ is that it diagonalize the Hamiltonian in the subspace of $q\beta$, there is freedom in the choice of this state. We can exploit this freedom and choose the $|\phi_0^i\rangle$ to be an eigenstates of $\hat{\omega}_i^{1,0}$ corresponding to real eigenvalues $\omega_i^{1,0}$:

$$\left[\mathcal{H}^d + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i(\omega_i)\rangle = \left(\omega_i^1 - \mathcal{H}^d \right) |1\rangle |\phi_1^i\rangle + \left(\omega_i^0 - \mathcal{H}^d \right) |0\rangle |\phi_0^i\rangle \quad (1.11)$$

If we now substitute the expansion 1.6 and gather the terms that result in $\hat{n}_{q\beta} = 1$, we get

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = (\omega_i^1 - \mathcal{H}^d) |1\rangle |\phi_1^i\rangle \quad (1.12)$$

Similarly, gathering the terms that result in $\hat{n}_{q\beta} = 0$ gives

$$T^\dagger c_{q\beta} |1\rangle |\phi_1^i\rangle = (\omega_i^0 - \mathcal{H}^d) |0\rangle |\phi_0^i\rangle \quad (1.13)$$

We now define two many-particle transition operators:

$$\begin{aligned} \eta^\dagger(\omega_i^1) &= \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T \equiv G_1 c_{q\beta}^\dagger T \\ \eta(\omega_i^0) &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_{q\beta} \equiv G_0 T^\dagger c_{q\beta} \end{aligned} \quad (1.14)$$

where G_j is the propagator $\frac{1}{\omega_i^j - \mathcal{H}^d}$. We can write this compactly as

$$\eta(\hat{\omega}) = G T^\dagger c_{q\beta} = \frac{1}{\hat{\omega} - \mathcal{H}^d} T^\dagger c_{q\beta} \quad (1.15)$$

where $\hat{\omega}_i = \omega_i^0 (1 - \hat{n}_{q\beta}) + \omega_i^1 \hat{n}_{q\beta} = \begin{pmatrix} \omega_i^1 & \\ & \omega_i^0 \end{pmatrix}$ is a 2x2 matrix and $\mathcal{H}^d = \mathcal{H}_0^d (1 - \hat{n}_{q\beta}) + \mathcal{H}_1^d \hat{n}_{q\beta}$ and $G = (\hat{\omega} - \mathcal{H}^d)^{-1}$. It is easy to check that this reproduces the previous forms of η_0 and $\eta_1^\dagger$. We will later find that it is important to demand that these two be Hermitian conjugates of each other; that constraint is imposed on the denominators:

$$\eta^\dagger(\omega_i^0) = \eta^\dagger(\omega_i^1) \implies \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T = c_{q\beta}^\dagger T \frac{1}{\omega_i^0 - \mathcal{H}^d} \quad (1.16)$$

Henceforth we will assume that this constraint has been imposed.

In terms of these operators, eq. 1.13 becomes

$$|1\rangle |\phi_1^i\rangle = \eta^\dagger |0\rangle |\phi_0^i\rangle, \quad |0\rangle |\phi_0^i\rangle = \eta |1\rangle |\phi_1^i\rangle \quad (1.17)$$

These allow us to write

$$|\psi_1\rangle = |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle = (1 + \eta) |1\rangle |\phi_1^i\rangle, \quad |\psi_0\rangle = (1 + \eta^\dagger) |0\rangle |\phi_0^i\rangle \quad (1.18)$$

Recalling that $|\psi_i\rangle = U^\dagger |i\rangle$, we can read off the required transformation:

$$U_1 = 1 + \eta \quad (1.19)$$

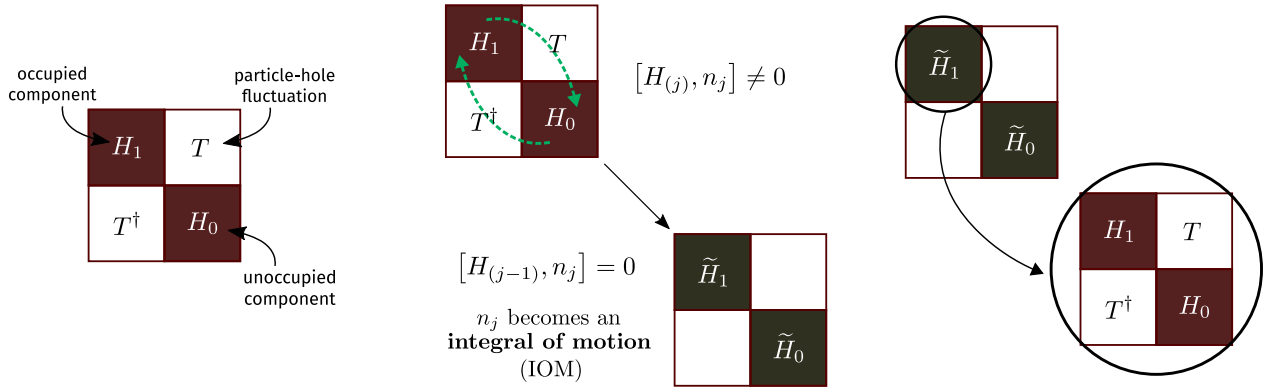


Figure 1.1: Three steps of the URG: Decompose the Hamiltonian in a 2×2 matrix, apply the unitary operator to rotate it, then repeat these steps with one of the rotated blocks.

Properties of the many-body transition operators

The operators η have some important properties. First is the Fermionic nature:

$$\eta^2 = \eta^{\dagger 2} = 0 \quad [c^{\dagger 2} = c^2 = 0] \quad (1.20)$$

Second is:

$$\begin{aligned} |1\rangle |\phi_1^i\rangle &= \eta^\dagger |0\rangle |\phi_0^i\rangle = \eta^\dagger \eta |1\rangle |\phi_1^i\rangle \implies \eta^\dagger \eta = \hat{n}_{q\beta} \\ |0\rangle |\phi_0^i\rangle &= \eta |1\rangle |\phi_1^i\rangle = \eta \eta^\dagger |0\rangle |\phi_0^i\rangle \implies \eta \eta^\dagger = 1 - \hat{n}_{q\beta} \end{aligned} \quad (1.21)$$

and hence the anticommutator

$$\implies \{\eta, \eta^\dagger\} = 1 \quad (1.22)$$

Note that the three equations in 1.21 work only when applied on the eigenstate $|\psi_i\rangle$ and not any arbitrary state.

$$\begin{aligned} \eta^\dagger \eta |\psi_i\rangle &= |1\rangle |\phi_1^i\rangle = \hat{n}_{q\beta} |\psi_i\rangle \\ \eta \eta^\dagger |\psi_i\rangle &= |0\rangle |\phi_0^i\rangle = (1 - \hat{n}_{q\beta}) |\psi_i\rangle \\ \{\eta^\dagger, \eta\} |\psi_i\rangle &= |\psi_i\rangle \end{aligned}$$

Form of the unitary operators

Although we have found the correct similarity transformations U_i (eqs. 1.19), we need to convert them into a unitary transformation. Say we are trying to rotate the eigenstate $|\psi_1\rangle$ into the state $|1\rangle$. We can then work with the transformation

$$U_1 = 1 + \eta \quad (1.23)$$

In this form, this transformation is not unitary. It can however be written in an exponential form:

$$U_1 = e^\eta \quad (1.24)$$

using the fact that $\eta^2 = 0$. It is shown in ref. [5] that corresponding to a similarity transformation e^ω , there exists a unitary transformation e^G where

$$G = \tanh^{-1} (\omega - \omega^\dagger) \quad (1.25)$$

Applying that to the problem at hand gives

$$U_1^\dagger = \exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} \quad (1.26)$$

Let $x = \tanh y$. Then,

$$x = \frac{e^{2y} + 1}{e^{2y} - 1} \implies y = \frac{1}{2} \log \frac{1+x}{1-x} \implies e^y = e^{\tanh^{-1} x} = \sqrt{\frac{1+x}{1-x}} \quad (1.27)$$

Therefore,

$$\exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} = \frac{1 + \eta - \eta^\dagger}{\sqrt{(1 + \eta^\dagger - \eta)(1 - \eta^\dagger + \eta)}} = \frac{1 + \eta - \eta^\dagger}{\sqrt{1 + \{\eta, \eta^\dagger\}}} = \frac{1}{\sqrt{2}} (1 + \eta - \eta^\dagger) \quad (1.28)$$

The *unitary* operator that transforms the entangled eigenstate $|\psi_1\rangle$ to the state $|1\rangle$ is thus

$$U_1 = \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \quad (1.29)$$

It can also be written as $\exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\}$ because

$$\begin{aligned} \exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\} &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} + \frac{1}{2!} (\eta^\dagger - \eta)^2 \left(\frac{\pi}{4} \right)^2 + \frac{1}{3!} (\eta^\dagger - \eta)^3 \left(\frac{\pi}{4} \right)^3 + \dots \\ &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} - \frac{1}{2!} \left(\frac{\pi}{4} \right)^2 - \frac{1}{3!} (\eta^\dagger - \eta) \left(\frac{\pi}{4} \right)^3 + \frac{1}{4!} \left(\frac{\pi}{4} \right)^4 + \dots \\ &= \cos \frac{\pi}{4} + (\eta^\dagger - \eta) \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \end{aligned} \quad (1.30)$$

There we used

$$(\eta^\dagger - \eta)^2 = \eta^{\dagger 2} + \eta^2 - \{\eta^\dagger, \eta\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0 \right] \quad (1.31)$$

and hence

$$(\eta^\dagger - \eta)^3 = -1 (\eta^\dagger - \eta) \quad (1.32)$$

and so on.

Effective Hamiltonian

We can now compute the form of the effective Hamiltonian that comes about when we apply U_1 - that is - when we rotate one exact eigenstate $|\psi_1\rangle$ into the occupied Fock space basis $|1\rangle$. From eq. 1.29,

$$\begin{aligned}
 U_1 \mathcal{H} U_1^\dagger &= \frac{1}{2} \left(1 + \eta^\dagger - \eta \right) \mathcal{H} \left(1 + \eta - \eta^\dagger \right) \\
 &= \frac{1}{2} \left(1 + \eta^\dagger - \eta \right) \left(\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right)
 \end{aligned} \tag{1.33}$$

In the last two lines, we expanded the Hamiltonian into the three parts $\mathcal{H}^d$, $\mathcal{H}^i$ and a third piece $\mathcal{H}^l \equiv c_{q\beta}^\dagger T + T^\dagger c_{q\beta}$.

For reasons that will become apparent, we will split the terms into two groups:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\underbrace{\mathcal{H}^d + \mathcal{H}^i + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger}_{\text{group 1}} + \overbrace{\mathcal{H}^l - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta}^{\text{group 2}} \right) \tag{1.34}$$

Group 2 can be easily shown to be 0. Note that terms that have two η or two $\eta^\dagger$ sandwiching a $\mathcal{H}$ can only be nonzero if the intervening $\mathcal{H}$ has an odd number of creation or destruction operators.

$$\eta \mathcal{H} \eta = \eta c_q^\dagger T \eta \tag{1.35}$$

and

$$\eta^\dagger \mathcal{H} \eta^\dagger = \eta^\dagger T^\dagger c_q \eta^\dagger \tag{1.36}$$

Group 2 becomes

$$\text{group 2} = \mathcal{H}^l - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta = c_q^\dagger T + T^\dagger c_q - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta \tag{1.37}$$

To simplify this, we use the relation

$$\begin{aligned}
 \eta c_q^\dagger T \eta &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_q c_q^\dagger T \eta \\
 &= T^\dagger c_q \frac{1}{\omega_i^1 - \mathcal{H}^d} c_q^\dagger T \eta \quad [\text{eq. 1.16}] \\
 &= T^\dagger c_q \eta^\dagger \eta \quad [\text{eq. 1.15}] \\
 &= T^\dagger c_q \hat{n}_q \quad [\text{eq. 1.21}]
 \end{aligned} \tag{1.38}$$

which gives

$$\eta c_q^\dagger T \eta = T^\dagger c_q \quad (1.39)$$

Taking the Hermitian conjugate of eq. 1.39 gives

$$\eta^\dagger T^\dagger c_q \eta^\dagger = c_q^\dagger T \quad (1.40)$$

Substituting the expressions 1.39 and 1.40 into the expression for group 2, 1.37, shows that it vanishes. This leaves us only with group 1:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \overbrace{\eta^\dagger \mathcal{H} \eta + \eta \mathcal{H} \eta^\dagger}^{\text{group A}} + \underbrace{[\eta^\dagger - \eta, \mathcal{H}]}_{\text{group B}} \right) \quad (1.41)$$

Group A simplifies in the following way. First note that $\eta^\dagger \mathcal{H}^I \eta = \eta^\dagger \mathcal{H}^I \eta = 0$ must be 0 because it will involve consecutive $c_{q\beta}$ or consecutive $c_{q\beta}^\dagger$. We are therefore left with the diagonal part of $\mathcal{H}$, which is $H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})$.

$$\eta^\dagger [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta + \eta [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta^\dagger = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger \quad (1.42)$$

This can be shown to be equal to the diagonal part:

$$\text{group A} = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i \quad (1.43)$$

It can also be shown that

$$\text{group B} = [\eta^\dagger - \eta, \mathcal{H}] = 2 [c_{q\beta}^\dagger T, \eta] \quad (1.44)$$

Putting it all together,

$$\tilde{\mathcal{H}} = \mathcal{H}^d + \mathcal{H}^i + [c_{q\beta}^\dagger T, \eta] \quad (1.45)$$

The renormalizing in the Hamiltonian is

$$\Delta \mathcal{H} = \tilde{\mathcal{H}} - \mathcal{H}^d - \mathcal{H}^i = [c_{q\beta}^\dagger T, \eta] \quad (1.46)$$

Because of eq. 1.44, it can also be written as

$$\Delta \mathcal{H} = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}_X] = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}] \quad (1.47)$$

This form will be useful later when we make the connection with one-shot Schrieffer-Wolff transformation and CUT RG.

To check that the renormalised Hamiltonian indeed commutes with $\hat{n}_{q\beta}$,

$$\begin{aligned} [\tilde{\mathcal{H}}, \hat{n}_{q\beta}] &= [[c_{q\beta}^\dagger T, \eta], \hat{n}_{q\beta}] = [c_{q\beta}^\dagger T \eta, \hat{n}_{q\beta}] - [\eta c_{q\beta}^\dagger T, \hat{n}_{q\beta}] \\ &= c_{q\beta}^\dagger T \eta \hat{n}_{q\beta} - \hat{n}_{q\beta} c_{q\beta}^\dagger T \eta \quad \left[2^{\text{nd}} [\cdot] \text{ is 0, } \because c_{q\beta}^\dagger \hat{n}_{q\beta} = \hat{n}_{q\beta} \eta = 0 \right] \\ &= c_{q\beta}^\dagger T \eta - c_{q\beta}^\dagger T \eta = 0 \end{aligned} \quad (1.48)$$

Fixed point condition

Within the URG, it is a prescription that the fixed point is reached when the denominator of the RG equation vanishes. This is equivalent to either $\omega_i^1 = \mathcal{H}_1^d$ or $\omega_i^0 = \mathcal{H}_0^d$. This shows that at the fixed point, one of the eigenvalues of $\hat{\omega}_i$ matches the corresponding eigenvalue of the diagonal blocks. This also leads to the vanishing of the off-diagonal block, because eqs. 1.12 and 1.13 gives

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = \left(\omega_i^1 - \mathcal{H}_1^d \right) |1\rangle |\phi_1^i\rangle = 0 \implies c_{q\beta}^\dagger T = 0 \quad (1.49)$$

Multiple off-diagonal terms

There is a subtle assumption in the definitions eq. 1.14. In order for η to be the Hermitian conjugate of $\eta^\dagger$, $\mathcal{H}_d$ cannot have any information that relates to the structure of T . To see why, say the total off-diagonal term is composed of two parts: $T = T_1 + T_2$.

$$\begin{aligned} \eta &= \frac{1}{\omega_0 - \mathcal{H}_d} \left(T_1^\dagger + T_2^\dagger \right) c = \left[\frac{1}{\omega^0 - E_1^0} T_1^\dagger c + \frac{1}{\omega^0 - E_2^0} T_2^\dagger c \right] \\ \eta^\dagger &= \frac{1}{\omega^1 - \mathcal{H}_d} c^\dagger (T_1 + T_2) = \left[\frac{1}{\omega^1 - E_1^1} c^\dagger T_1 + \frac{1}{\omega^1 - E_2^1} c^\dagger T_2 \right] \end{aligned} \quad (1.50)$$

where $\mathcal{H}_d T_i^\dagger c = E_i^0 T_i^\dagger c$ and $\mathcal{H}_d c^\dagger T_i = E_i^1 c^\dagger T_i$. We can now see that in order for $\eta = (\eta^\dagger)^\dagger$ to hold, two conditions must be met:

$$\omega^0 - E_1^0 = \omega^1 - E_1^1, \quad \omega^0 - E_2^0 = \omega^1 - E_2^1 \quad (1.51)$$

This will not hold generally. The correct solution is to realize that each such off-diagonal term T_i will come with its own quantum fluctuation scale ω_i .

$$\begin{aligned} \eta &= \sum_i \frac{1}{\omega_i^0 - E_i^0} T_i^\dagger c \\ \eta^\dagger &= \sum_i \frac{1}{\omega_i^1 - E_i^1} c^\dagger T_i \end{aligned} \quad (1.52)$$

If we now impose the condition that $\eta = (\eta^\dagger)^\dagger$, we get the relations

$$\omega_i^0 - \omega_i^1 = E_i^0 - E_i^1 \quad (1.53)$$

and so

$$\eta^\dagger - \eta = \sum_i \frac{1}{\omega_i^0 - E_i^0} \left(c^\dagger T_i - T_i^\dagger c \right) \quad (1.54)$$

The expression for the renormalization will not be just $[c^\dagger T, \eta]$ in this case. That form will be non-Hermitian. The correct form is obtained from the more general form $[\eta^\dagger - \eta, \mathcal{H}_X]$:

$$\begin{aligned}\Delta\mathcal{H} &= \frac{1}{2} [\eta^\dagger - \eta, c^\dagger T + T^\dagger c] = \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} [c^\dagger T_i - T_i^\dagger c, c^\dagger T_j + T_j^\dagger c] \\ &= \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} \left[\hat{n} (T_i T_j^\dagger + T_j T_i^\dagger) - (1 - \hat{n}) (T_i^\dagger T_j + T_j^\dagger T_i) \right] \\ &= \frac{1}{2} \sum_{ij} \left(\frac{1}{\omega_i^0 - E_i^0} + \frac{1}{\omega_j^0 - E_j^0} \right) [\hat{n} T_i T_j^\dagger - (1 - \hat{n}) T_i^\dagger T_j]\end{aligned}\quad (1.55)$$

Prescription

Given a Hamiltonian

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_0 + c^\dagger T + T^\dagger c \quad (1.56)$$

the goal is to look at the renormalization of the various couplings in the Hamiltonian as we decouple high energy electron states. Typically we have a shell of electrons at some energy D . During the process, we make one simplification. We assume that there is only one electron on that shell at a time, say with quantum numbers q, σ , and calculate the renormalization of the various couplings due to this electron. We then sum the momentum q over the shell and the spin β , and this gives the total renormalization due to decoupling the entire shell.

From eq. 1.45, the first two terms in the rotated Hamiltonian are just the diagonal parts of the bare Hamiltonian; they are unchanged in that part. The renormalization comes from the third term. For one electron $q\beta$ on the shell, the renormalization is

$$\Delta\mathcal{H} = [c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}), \eta] = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta - \eta c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.57)$$

Since this assumes we have obtained this from U_1 , it is fair to tag the η with a suitable label:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 - \eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.58)$$

It is clear that the first term takes into account virtual excitations that start from a filled state ($\hat{n}_{q\beta} = 1$ initially) - such a term is said to be a part of the *particle sector*.

$$\Delta_1 \mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 \quad (1.59)$$

The second term, on the other hand, considers excitations that start from an empty state. They constitute the *hole sector*.

$$\Delta_0 \mathcal{H} = -\eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.60)$$

To write the total renormalization in a particle-hole symmetric form, we can use the relation $\eta_0 = -\eta_1$, such that both the terms will now come with a positive sign:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 + \eta_0 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.61)$$

We can make one more manipulation: using eq. 1.16, we get

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \eta_1 + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \eta_0^\dagger \quad (1.62)$$

This form of the total renormalization is identical to the one we use in the "Poor Man's scaling"-type of renormalization that was used to get the scaling equations in the Kondo and Anderson models [6, 7]. Writing down the forms of η and $\eta^\dagger$ explicitly, we get

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \frac{1}{\omega_0^1 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \quad (1.63)$$

The renormalization due to the entire shell is obtained by summing over all states on the shell.

$$\Delta\mathcal{H} = \sum_{q\beta} \left[c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \frac{1}{\omega_0^1 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \right] \quad (1.64)$$

These equations will now need to be simplified. For example, in the particle sector, we can set $\hat{n}_{q\beta} = 0$ in the numerator, because there is no such excitation in the initial state. Similarly, in the hole sector, we can set $\hat{n}_{q\beta} = 1$ because that state was occupied in the initial state. Another simplification we typically employ is that $\mathcal{H}_{0,1}^d$ will, in general, have the energies of all the electrons. But we consider only the energy of the on-shell electrons in the denominator. After integrating out these electrons, we can rearrange the remaining operators to determine which term in the Hamiltonian it renormalizes and what is the renormalization.

At first sight, one might think that we must evaluate lots of traces to obtain the terms in $\Delta\mathcal{H}$. A little thought reveals that the terms in the numerator are simply the off-diagonal terms in the Hamiltonian; $\text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta}$ is the off-diagonal term that has $c_{q\beta}$ in it, and $c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta})$ is the off-diagonal term that has $c_{q\beta}^\dagger$ in it. $\mathcal{H}^D$ is just the diagonal part of the Hamiltonian.

1.2 Example: Unitary RG analysis of the Kondo model

We now describe the implementation of the unitary RG method in a simple paradigmatic model, the single-channel Kondo model. It describes the scattering of itinerant non-interacting electrons off a magnetic impurity $\mathbf{S}_d$, and is defined by the Hamiltonian

$$H_K = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \mathbf{S}_d \cdot \frac{1}{2} \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k_1,k_2} c_{k_1,\alpha}^\dagger c_{k_2,\beta} \cdot \quad (2.1)$$

Note that the Kondo term is effectively of the form $\mathbf{S}_1 \cdot \mathbf{S}_2$ with the local spin density of the conduction bath making up the second spin:

$$\frac{1}{2} \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k_1,k_2} c_{k_1,\alpha}^\dagger c_{k_2,\beta} = \frac{1}{2} \sum_{\alpha,\beta} \sigma_{\alpha,\beta} c_{0,\alpha}^\dagger c_{0,\beta} = \mathbf{s}_0 \cdot \quad (2.2)$$

As a result, the Kondo term can be written in the form

$$S_d^z s_0^z + \frac{1}{2} S_d^+ s_0^- + \frac{1}{2} S_d^- s_0^+ . \quad (2.3)$$

We will make use of this form.

Because of spin-flip scattering processes, the problem is manybody in nature. In order to obtain the low-energy physics, we will progressively integrate out high-energy modes in the conduction bath and approach the low-energy degrees of freedom. Such an approach is needed because the high and low-energy states interact with each other through the impurity spin and cannot be trivially removed from the problem. The way this works is that the process of integrating out high-energy states generates additional scattering processes within the low-energy states; if these processes already exist in the Hamiltonian, the new contributions *renormalise* the existing couplings, but if the processes don't exist, then the Hamiltonian generates new terms as low energies are approached.

Consider an arbitrary j^{th} step of the process where we wish to integrate out the conduction bath states within a narrow shell about the energies $\pm E_j$ and retain only the states at $|E| < |E_j|$. We designate the states on this shell by the momentum index q , and the states below it by indices k_1 and k_2 . We need to first separate the Hamiltonian into diagonal and off-diagonal terms from the perspective of the shell being integrated out:

$$\begin{aligned} H_D &= \sum_{q,\sigma} \epsilon_q n_{q,\sigma} + JS_d^z \frac{1}{2} \sum_q (n_{q,\uparrow} - n_{q,\downarrow}) , \\ H_X &= JS_d \cdot \frac{1}{2} \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k,q} \left(c_{k,\alpha}^\dagger c_{q,\beta} + c_{q,\alpha}^\dagger c_{k,\beta} \right) . \end{aligned} \quad (2.4)$$

In order to make the particle-hole symmetry in the kinetic energy term manifest, we replace the number operators with pseudospin operators $\tau = n - 1/2$ and drop the constant term:

$$\begin{aligned} H_D &= E_j \sum_{q,\sigma} \tau_{q,\sigma} + JS_d^z \sum_q (n_{q,\uparrow} - n_{q,\downarrow}) , \\ H_X &= JS_d \cdot \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k,q} \left(c_{k,\alpha}^\dagger c_{q,\beta} + c_{q,\alpha}^\dagger c_{k,\beta} \right) . \end{aligned} \quad (2.5)$$

This ensures that particle and hole excitations both cost the same energy. We now describe how to obtain the renormalisation in the Hamiltonian arising from unitarily decoupling the states on the shell. As mentioned earlier, this happens because the process of integrating out the states q generates interactions between the other states $\{k\}$ and this modifies the Hamiltonian couplings.

To keep things brief, we focus on obtaining the renormalisation of the following spin-flip term in the Kondo Hamiltonian:

$$\frac{1}{2} JS_d^+ \sum_{k_1, k_2} c_{k_1, \downarrow}^\dagger c_{k_2, \uparrow} \quad (2.6)$$

For that, we need to figure out which combinations of off-diagonal terms in H_X can result in such terms, when the q -states are integrated out. One possibility is

$$\frac{1}{4} \sum_{q, k_1, k_2} JS_d^z c_{q, \uparrow}^\dagger c_{k_2, \uparrow} \frac{1}{\omega - H_D} JS_d^+ c_{k_1, \downarrow}^\dagger c_{q, \uparrow} . \quad (2.7)$$

In order to evaluate H_D in the denominator, we note that the hole excitation of $q \uparrow$ at the right (i) indicates that ϵ_q is positive, and the configuration $n_{q,\uparrow} = 0$ in the intermediate state means that $\epsilon_q \tau_{q,\uparrow}$ works out to be $|E_j|/2$, and (ii) the impurity and bath spins are antiparallel in the intermediate state because of the opposite spins of their excitations, and so the J -term in the denominator evaluates to $J(1/2)(-1/2) = -J/4$. Substituting these in the expression (and noting the minus sign in front of H_D) gives

$$\frac{1}{4} J S_d^z c_{q,\uparrow}^\dagger c_{k_2,\uparrow} \frac{1}{\omega - |E_j|/2 + J/4} J S_d^+ c_{k_1,\downarrow}^\dagger c_{q,\uparrow} = - \frac{J^2/4}{\omega - |E_j|/2 + J/4} S_d^z c_{k_2,\downarrow}^\dagger c_{k_1,\uparrow} n_{q,\uparrow} . \quad (2.8)$$

In order to truly integrate out the mode q , we set $n_{q,\uparrow} = 1$, guided by the fact that the mode was initially occupied.

Additional processes can be constructed by allowing the initial excitation to be of the particle kind:

$$\sum_{q,k_1,k_2} J S_d^z c_{k_1,\uparrow}^\dagger c_{q,\uparrow} \frac{1}{\omega - H_D} J S_d^+ c_{q,\downarrow}^\dagger c_{k_2,\uparrow} . \quad (2.9)$$

1.3 Renormalised perturbation theory about fixed points

In this section, we will describe the technique of carrying out renormalised theory about a gapless fixed point theory towards obtaining a low-energy effective Hamiltonian for the excitations above the ground state. This is useful for characterising metallic phases, and detecting whether the metal is, say, a Landau Fermi liquid or a non-Fermi liquid.

We consider here the case of the single-channel Kondo model. We would like to demonstrate the method by deriving a local Fermi liquid theory for the gapless excitations above the ground state, as obtained by Nozières [8]. In its simplest form, the Kondo model features a spin-1/2 local moment undergoing antiferromagnetic spin-exchange scattering with the local spin-density of a non-interacting electronic conduction bath:

$$H_K = J \mathbf{S}_d \cdot \mathbf{s} + \sum_{k,\sigma} \epsilon_k n_{k,\sigma} , \quad (3.1)$$

where S_d and s are the spins of the impurity and the conduction bath. For any non-zero value of the Kondo coupling J (the coupling for the impurity-bath spin exchange processes), it can be shown that J is relevant under a renormalisation group flow from high energy to low-energy, and the coupling effectively diverges as the fixed point is approached [9]. At the fixed point, the kinetic energy of the conduction bath can be ignored relative to the Kondo coupling, and the ground state is a spin-singlet between the impurity $\mathbf{S}_d$ and the conduction bath local spin density $\mathbf{s}$:

$$|\Psi\rangle_0 = \frac{1}{\sqrt{2}} (|\uparrow_d, \Downarrow\rangle - |\downarrow_d, \Uparrow\rangle) , \quad (3.2)$$

where $\uparrow_d$ represents the configuration of the impurity and the thick arrow is the configuration of the conduction bath spin.

In order to obtain the excitations, we now note that because of the RG-relevance of J , the first term in the Hamiltonian in eq. 3.1 can only have finite energy excitations. Since we are looking for excitations that cost vanishingly small energy, we focus on the second term. With an eye towards developing a perturbation theory over $1/J$, we recast the kinetic energy term in real-space as a tight-binding Hamiltonian connected to the local density that hybridises with the impurity:

$$H_{\text{KE}} = -t^* \sum_{i,\sigma} (c_{i,\sigma}^\dagger c_{i+1,\sigma} + \text{h.c.}) . \quad (3.3)$$

Hopping processes from the zeroth site modify the singlet ground state and lead to virtual excitations. We will now obtain the effect of this Hamiltonian by obtaining its renormalisation effects within the ground state subspace.

To better frame the perturbation theory which we will use to obtain the renormalisation effects, we clarify the zeroth Hamiltonian H_0 and perturbation V :

$$\begin{aligned} H_0 &= J^* \mathbf{S}_d \cdot \mathbf{s}, \\ V &= -t^* \sum_{i,\sigma} (c_{i,\sigma}^\dagger c_{i+1,\sigma} + \text{h.c.}) . \end{aligned} \quad (3.4)$$

The goal is to obtain the shifts in the energies of H_0 arising from the perturbation term. The eigenstates of H_0 are

$$\begin{aligned} |\Psi\rangle_{0,0,2} &= \frac{1}{\sqrt{2}} (|\uparrow_d, \downarrow\rangle - |\downarrow_d, \uparrow\rangle) : & E &= -3J^*/4 \\ |\Psi\rangle_{\frac{1}{2}, \frac{\sigma}{2}, 1} &= |\sigma_d, 0\rangle : & E &= 0 \\ |\Psi\rangle_{\frac{1}{2}, \frac{\sigma}{2}, 3} &= |\sigma_d, 2\rangle : & E &= 0 \\ |\Psi\rangle_{1,0,2} &= \frac{1}{\sqrt{2}} (|\uparrow_d, \downarrow\rangle + |\downarrow_d, \uparrow\rangle) : & E &= J^*/4 \\ |\Psi\rangle_{1,\sigma,2} &= |\sigma_d, \sigma\rangle : & E &= J^*/4, \end{aligned} \quad (3.5)$$

where the quantum numbers in the subscript are the total spin $S = S_d + s$, the z -component of the total spin $S^z = S_d^z + s^z$, and the total occupancy $N = n_d + n$. The second, third and fifth states are doubly degenerate with σ taking values ± 1 . The ground state is of course $|\Psi\rangle_{0,0,2}$. However, this only takes into account the zeroth site in the conduction bath. In order to consider perturbative processes where an electron hops from the zeroth site to the next (first) site, we need to involve the first site as well. Since the first site does not have any energy in H_0 , the eigenstates simply become four-fold degenerate, due to the four possible configurations of the first site. The eigenstates are then labelled as $\Psi_{S,S^z,N,c_1} = |\Psi_{S,S^z,N}\rangle \otimes |c_1\rangle$ where c_1 can be 0, $\uparrow$, $\downarrow$ or 2 depending on the configuration of the first site.

To obtain the excitations, we will calculate the perturbative shifts in the four-fold degenerate ground state subspace spanned by the states $\{\Psi_{0,0,2,0}, \Psi_{0,0,2,\uparrow}, \Psi_{0,0,2,\downarrow}, \Psi_{0,0,2,2}\}$. We will do this order by order, stopping when we obtain the lowest order non-trivial Hamiltonian. At first and third

order, the energy shifts are given by

$$\begin{aligned}\Delta E_n^{(1)} &= \langle \Psi_n | V | \Psi_n \rangle, \\ \Delta E_n^{(3)} &= \sum_{k_2 \neq n, k_3 \neq n} \frac{\langle \Psi_n | V | \Psi_{k_3} \rangle \langle \Psi_{k_3} | V | \Psi_{k_2} \rangle \langle \Psi_{k_2} | V | \Psi_n \rangle}{(E_n - E_{k_3})(E_n - E_{k_2})} - \Delta E_n^{(1)} \sum_{k_3 \neq n} \frac{|\langle \Psi_n | V | \Psi_{k_3} \rangle|^2}{(E_n - E_{k_3})^2}.\end{aligned}\quad (3.6)$$

Both these contributions vanish because of mismatched quantum numbers: upon applying the perturbation Hamiltonian V any odd number of times, the final resultant state will have a different occupancy on the impurity+zeroth site compared to the starting state. Since H_0 conserves the occupancy of the impurity+zeroth site complex, the overlap between such states is zero. More specifically, what this means for the first term of $\Delta E_n^{(3)}$ is the following:

- the excited state k_2 will have either 1 or 3 electrons on the $d+0$ complex (because the ground state hosts 2 electrons there, and V will either add an electron there or remove one electron from it).
- Another application of V means that k_3 will have 0, 2 or 4 electrons.
- A final application of V means that the terminal state will have an odd number of electrons – 1, 3 or 5,

The overlap of this state with the ground state – $\langle \Psi_n | V | \Psi_{k_3} \rangle$ – will therefore vanish.

We therefore focus on the second and fourth order effects. At second order, the renormalisation is given by

$$\Delta E_n^{(2)} = \sum_{k_2 \neq n} \frac{|\langle \Psi_n | V | \Psi_{k_2} \rangle|^2}{E_n - E_{k_2}}. \quad (3.7)$$

We work with the shifts in the $(0, 0, 2, 0)$ and $(0, 0, 2, \uparrow)$ states, because the shifts of $(0, 0, 2)$ and $(0, 0, \downarrow)$ can be inferred, respectively, from them using particle-hole symmetry and spin-rotation symmetry. For the state $(0, 0, 2, 0)$, we have

$$V |\Psi_{0,0,2,0}\rangle = -t^* \frac{1}{\sqrt{2}} (|\uparrow_d, 0, \downarrow_1\rangle - |\downarrow_d, 0, \uparrow_1\rangle) = \frac{-t^*}{\sqrt{2}} |\Psi\rangle_{\frac{1}{2}, \frac{1}{2}, 1, \downarrow} + \frac{t^*}{\sqrt{2}} |\Psi\rangle_{\frac{1}{2}, -\frac{1}{2}, 1, \uparrow}. \quad (3.8)$$

Summing k_2 over the states $\frac{1}{2}, \frac{1}{2}, 1, \downarrow$ and $\frac{1}{2}, -\frac{1}{2}, 1, \uparrow$ with energies $E_n - E_{k_2} = -3J^*/4$, the total shift comes out to be

$$\Delta E_{(0,0,2,0)}^{(2)} = 2 \frac{t^{*2}}{2} \times \frac{1}{-3J^*/4} = -\frac{4t^{*2}}{3J^*}. \quad (3.9)$$

For the state $(0, 0, 2, \uparrow)$, we have

$$V |\Psi_{0,0,2,\uparrow}\rangle = t^* \frac{1}{\sqrt{2}} (|\uparrow_d, 0, 2\rangle + |\uparrow_d, 2, 0\rangle) = \frac{t^*}{\sqrt{2}} |\Psi\rangle_{\frac{1}{2}, \frac{1}{2}, 1, 2} + \frac{t^*}{\sqrt{2}} |\Psi\rangle_{\frac{1}{2}, \frac{1}{2}, 1, 0}. \quad (3.10)$$

Summing k_2 over the appropriate eigenstates and using $E_n - E_{k_2} = -3J^*/4$ gives the same shift:

$$\Delta E_{(0,0,2,\uparrow)}^{(2)} = -\frac{4t^{*2}}{3J^*}. \quad (3.11)$$

Using symmetries, we also have

$$\Delta E_{(0,0,2,\downarrow)}^{(2)} = \Delta E_{(0,0,2,2)}^{(2)} = -\frac{4t^{*2}}{3J^*} . \quad (3.12)$$

We also note that off-diagonal matrix elements are zero in such a setup, because any two states in the ground state subspace will differ by at least one of the two conserved quantum numbers s_1^z and n_1 .

We now need to convert these shifts into a renormalised Hamiltonian that describes the excitations arising from such virtual scattering processes. In the absence of shift of any off-diagonal matrix element, the diagonal shifts will only lead to the generation of number-conserving operators. For example, the shift $\Delta E_{(0,0,2,0)}^{(2)} = -\frac{4t^{*2}}{3J^*}$ means that the energy of the null-occupied state of the first site is lowered by that amount. This can be represented through the operator

$$-\frac{4t^{*2}}{3J^*}(1 - n_{1\uparrow})(1 - n_{1\downarrow}) . \quad (3.13)$$

Writing down operator forms for the other shifts as well, we get

$$\Delta H^{(2)} = -\frac{4t^{*2}}{3J^*} [(1 - n_{1\uparrow})(1 - n_{1\downarrow}) + n_{1\uparrow}(1 - n_{1\downarrow}) + (1 - n_{1\uparrow})n_{1\downarrow} + n_{1\uparrow}n_{1\downarrow}] . \quad (3.14)$$

However, because the shifts in all the states are identical, adding up all the operators reduces the renormalisation to a constant shift:

$$\Delta H^{(2)} = -\frac{4t^{*2}}{3J^*} . \quad (3.15)$$

So we conclude that there's no non-trivial renormalisation at second order.

We now extend the perturbation theory to fourth order. Since the first and third order corrections vanish, and the second order correction only produces a constant shift, the first potentially non-trivial renormalisation arises at fourth order.

For a non-degenerate state, the fourth-order energy shift is

$$\Delta E_n^{(4)} = \sum_{k_1, k_2, k_3 \neq n} \frac{\langle \Psi_n | V | \Psi_{k_3} \rangle \langle \Psi_{k_3} | V | \Psi_{k_2} \rangle \langle \Psi_{k_2} | V | \Psi_{k_1} \rangle \langle \Psi_{k_1} | V | \Psi_n \rangle}{(E_n - E_{k_1})(E_n - E_{k_2})(E_n - E_{k_3})} - \Delta E_n^{(2)} \sum_{k_2 \neq n} \frac{|\langle \Psi_n | V | \Psi_{k_2} \rangle|^2}{(E_n - E_{k_2})^2} . \quad (3.16)$$

We first calculate it for the state $\Psi_{0,0,2,0}$, where the first site is unoccupied. We initially focus on the first term. One application of V gives

$$V |\Psi\rangle_{0,0,2,0} = -\frac{t}{\sqrt{2}} (|\uparrow_d, 0, \downarrow_1\rangle - |\downarrow_d, 0, \uparrow_1\rangle) . \quad (3.17)$$

This allows two possible states in k_2 : $|\uparrow_d, 0, \downarrow_1\rangle$ and $|\downarrow_d, 0, \uparrow_1\rangle$, with overlaps $\pm t/\sqrt{2}$. They are linked by particle-hole symmetry, so we will calculate the subsequent contribution from only the first and then double the result.

With $k_2 = |\uparrow_d, 0, \downarrow_1\rangle$, applying V once more gives

$$V |k_2\rangle = -t |\uparrow_d, \downarrow, 0_1\rangle . \quad (3.18)$$

This has overlap with the triplet state with zero spin, leading to $k_3 = \Psi_{1,0,2,0}$, with an overlap $-t/\sqrt{2}$. A third application of V gives

$$V|k_3\rangle = -\frac{t}{\sqrt{2}}(|\uparrow_d, 0, \downarrow_1\rangle + |\downarrow_d, 0, \uparrow_1\rangle) . \quad (3.19)$$

The final excited states k_4 are therefore the same as k_2 . A final application of V on the two possible k_4 gives

$$\begin{aligned} V|\uparrow_d, 0, \downarrow_1\rangle &= -t|\uparrow_d, \downarrow, 0_1\rangle , \\ V|\downarrow_d, 0, \uparrow_1\rangle &= -t|\downarrow_d, \uparrow, 0_1\rangle . \end{aligned} \quad (3.20)$$

The two final states have overlaps with opposite sign with the ground state. Since the excitation energies in the denominator in the first term of eq. 3.16 are the same for each of the paths, the sum over k_4 reduces to a sum over it's overlap with the ground state. As obtained above, this overlap is of different signs for the two possibilities of k_4 , leading to this vanishing of the first term of eq. 3.16.

For the second term, we have already calculated the overlap $\langle\Psi_n|V|k_2\rangle$ as well as the second order shift. Plugging those in gives

$$\Delta E_{0,0,2,0}^{(4)} = -\frac{4t^{*2}}{3J^*} \times 2\frac{t^{*2}/2}{(-3J^*/4)^2} = \frac{64t^{*4}}{27J^{*3}} . \quad (3.21)$$

This is the total fourth order shift for the state $\Psi_{0,0,2,0}$.

We next look at the state $\Psi_{0,0,2,\uparrow}$, starting with the first term. One application of the perturbation Hamiltonian gives

$$V|\Psi_{0,0,2,\uparrow}\rangle = \frac{-t^*}{\sqrt{2}}(-|\uparrow_d, 0, 2_1\rangle - |\uparrow, 2, 0_1\rangle) . \quad (3.22)$$

Proceeding exactly as before, the possible excited states k_2 are $|\uparrow_d, 0, 2_1\rangle$ and $|\uparrow, 2, 0_1\rangle$. The overlaps and denominators for both states are $t^*/\sqrt{2}$ and $-3J^*/4$ respectively. We now focus henceforth on the state $|\uparrow_d, 0, 2_1\rangle$ because the contribution from the other will be identical. Applying the perturbation on this state gives

$$V|\uparrow_d, 0, 2_1\rangle = -t^*(|\uparrow_d, \uparrow, \downarrow_1\rangle - |\uparrow_d, \downarrow, \uparrow_1\rangle) . \quad (3.23)$$

These gives two choices for k_3 : the $S^z = 1$ triplet state for $d + 0$ $|\uparrow_d, \uparrow, \downarrow_1\rangle$, and the $S^z = 0$ triplet state $\Psi_{1,0,2,\uparrow}$. While the denominators are $-J^*$ for both the states, the overlaps are $-t^*$ and $t^*/\sqrt{2}$ respectively.

Following the first state and applying V on it, we get

$$V|\uparrow_d, \uparrow, \downarrow_1\rangle = -t^*(|\uparrow, 0, 2_1\rangle + |\uparrow, 2, 0_1\rangle) . \quad (3.24)$$

The final set of excited states, k_4 , therefore work out to be $|\uparrow, 0, 2_1\rangle$ and $|\uparrow, 2, 0_1\rangle$, giving denominators $-3J^*/4$ and overlaps $-t^*$. The final application of V on either of these states gives $-t^*(|\uparrow_d, \uparrow, \downarrow_1\rangle - |\uparrow_d, \downarrow, \uparrow_1\rangle)$, and the final overlap with the ground state is $t^*/\sqrt{2}$. The total contribution from the excitation branch $\Psi_{0,0,2,\uparrow} \rightarrow |\uparrow_d, 0, 2_1\rangle \rightarrow |\uparrow_d, \uparrow, \downarrow_1\rangle \rightarrow \dots$ therefore comes out to be

$$S_1 = 2(t^*/\sqrt{2})(-t^*)^2(t^*/\sqrt{2})/((-3J^*/4)^2(-J^*)) = -\frac{16t^{*4}}{9J^{*3}} . \quad (3.25)$$

We now focus on the other excitation branch, $\Psi_{0,0,2,\uparrow} \rightarrow |\uparrow_d, 0, 2_1\rangle \rightarrow \Psi_{1,0,2,\uparrow} \rightarrow \dots$. Applying V on this excited state gives

$$V\Psi_{1,0,2,\uparrow} = \frac{t^*}{\sqrt{2}} (|\uparrow_d, 0, 2_1\rangle + |\uparrow_d, 2, 0_1\rangle) . \quad (3.26)$$

This gives the same k_3 as the contribution from the other branch, but with a reduced overlap $t^*/\sqrt{2}$ instead of t^* . Since we have already carried out the previous calculation, we can directly write down the result:

$$S_2 = S_1/2 = -\frac{8t^{*4}}{9J^{*3}} . \quad (3.27)$$

The factor of $1/2$ arises because the second and third overlaps are suppressed by $1/\sqrt{2}$ compared to the first branch S_1 . Combining S_1 and S_2 , the total contribution to the first term in the fourth order expression arising from the branch $\Psi_{0,0,2,\uparrow} \rightarrow |\uparrow_d, 0, 2_1\rangle \rightarrow \dots$ is

$$-\frac{8t^{*4}}{3J^{*3}} . \quad (3.28)$$

By particle-hole symmetry, the contribution from the other branch $\Psi_{0,0,2,\uparrow} \rightarrow |\uparrow_d, 2, 0_1\rangle \rightarrow \dots$ is identical. Together, the total fourth order renormalisation from the first term is

$$-\frac{16t^{*4}}{3J^{*3}} . \quad (3.29)$$

We now compute the second term in the fourth order expression. Again, the components have already been calculated during the second order calculation, so we just plug them in:

$$-\frac{-4t^{*2}}{3J^*} \times 2 \frac{t^{*2}/2}{(-3J^*/4)^2} = \frac{64t^{*4}}{27J^{*3}} . \quad (3.30)$$

Adding the two parts, we get for the total fourth order shift:

$$\Delta E_{0,0,2,\uparrow}^{(4)} = -\frac{80t^{*4}}{27J^{*3}} \quad (3.31)$$

Using particle-hole and spin rotation symmetries, we can now write down the fourth order shifts for all four states in the low-energy manifold:

$$\begin{aligned} \Delta E_{0,0,2,0}^{(4)} &= \Delta E_{0,0,2,2}^{(4)} = \frac{64t^{*4}}{27J^{*3}} \\ \Delta E_{0,0,2,\uparrow}^{(4)} &= \Delta E_{0,0,2,\downarrow}^{(4)} = -\frac{80t^{*4}}{27J^{*3}} . \end{aligned} \quad (3.32)$$

We can already see that the singly-occupied states experience a lowering of energy due to virtual excitations of the singlet state, while the doubly-occupied state has its energy raised. This effectively means if an electron is already present on the first site, a second incoming electron experiences a

repulsion due to it. Replacing each state with the appropriate operator, we can convert this intuition into an effective Hamiltonian for the excitations:

$$\begin{aligned}\Delta H^{(4)} &= \frac{16t^{*4}}{27J^{*3}} |\Psi_{0,0,2}\rangle \langle \Psi_{0,0,2}| \left[4(1 - n_{1,\uparrow})(1 - n_{1,\downarrow}) + 4n_{1,\uparrow}n_{1,\downarrow} - 5n_{1,\uparrow}(1 - n_{1,\downarrow}) - 5n_{1,\downarrow}(1 - n_{1,\uparrow}) \right] \\ &= \frac{16t^{*4}}{27J^{*3}} |\Psi_{0,0,2}\rangle \langle \Psi_{0,0,2}| \left[-u \sum_{\sigma} n_{1\sigma} + un_{1\uparrow}n_{1\downarrow} \right],\end{aligned}\tag{3.33}$$

where $u = 9$. This is the local Fermi liquid Hamiltonian for excitations above the Kondo screened ground state; the repulsive $n_{\uparrow}n_{\downarrow}$ term ensures that the excitations are electronic in nature.

1.4 Comparison of renormalisation group approaches based on canonical transformations

We have considered three canonical transformations in this section: the Poor Man's scaling (PMS), the Schrieffer-Wolff transformation (SWT) and the continuous unitary transformation renormalization group (CUT-RG). PMS and SWT are more or less identical; they differ in the context in which they are used. PMS is used when there is an entire spectrum of energies in the model, ranging from a low energy limit to a high energy; it is then employed to decouple the highest energy modes in an iterative fashion. SWT is used when the Hamiltonian can be split into one high and one low energy part, and we need to decouple these two modes. It is clear when seen from this perspective that SWT is like a one-shot PMS; it decouples the UV from the IR in one step, compared to the iterative approach of PMS. However, as has been shown in the previous subsections, both PMS and SWT can be formalized in an identical fashion, so that one can be switched for the other in both contexts.

Now that we have established that PMS and SWT are formally identical, we can relate them to URG. URG is similar in philosophy to PMS in the fact that URG also successively decouples high energy modes from the low energy ones. The difference is that URG takes care of the quantum fluctuations, at least in principle, by introducing a new set of energy scales ω . These ω then parameterise the RG flows of URG, compared to the single RG flow of PMS. Since SWT is formally the same as PMS, there is also a comparison between SWT and URG. Both PMS and SWT trivialize the quantum fluctuation scales of URG by replacing it with the bare energy values.

CUT-RG is philosophically different from the other transformations. Its goal is to gradually reduce the contributions of the off-diagonal terms by making them decay along a certain scale l . Off-diagonal terms that connect states with large energy differences decay faster. In this sense, there is no sequential dropping out of off-diagonal terms; all off-diagonal terms disappear at $l = \infty$. In this sense, it is like a continuous version of SWT. While SWT strives to remove an entire off-diagonal term in one-shot, CUT RG does this gradually by introducing a scale l . This separation of scales is absent in SWT. It does exist in URG and PMS, albeit in a different fashion. There, the separation comes in when we decouple single electron states starting from the Brillouin zone boundary Λ_N and come down to the Fermi surface Λ_0 . Each Λ provides a natural energy scale for separating the high and low energy physics.

If one integrates the continuum generator η over all the scales, one should recover something analogous to the SWT generator. This generator is however necessarily perturbative in the off-diagonal term, since by definition η will only be of first order in $\mathcal{H}_X$. This is in contrast to the non-perturbative generators in URG and, at least in principle, PMS. This non-perturbative form is encapsulated in the presence of a second completely different set of energy scales ω (or E in PMS). This second energy scale is absent in CUT RG because it takes care of at most the second order term.

Another point to note is that since SWT keeps the entire off-diagonal piece in the generator, new terms will almost certainly be generated at every step, and they have to be truncated. This is not the case with URG or PMS, because in those methods, we decouple just a single-electron at each step, and so those electrons become integrals of motion at that step, leading to their removal from the off-diagonal piece, and very often the Hamiltonian retains the same form as the bare model. This makes the philosophy of URG and PMS easier to work with. Tied to this is the fact that CUT RG often takes a certain type of interaction in the generator part, and not the entire off-diagonal piece. Hence, at the limit of the flow parameter going to ∞ , the chosen off-diagonal piece goes to zero but the remaining off-diagonal pieces still remain. As a result, the Hamiltonian is at most block diagonal at this stage. On the other hand, URG progressively decouples single electrons, meaning all scattering terms corresponding to that electron vanish at each step.

One last thing to note is that URG, being unitarily implemented with a well-defined generator, does not accommodate for spontaneous symmetry breaking (SSB). In order to see SSB, the symmetry-breaking term has to be added to the bare model explicitly; if this term grows under the RG, then the ground state will be symmetry-broken. CUT RG, however, does allow for SSB through the idea that the generator is not uniquely defined. If the generator commutes with a particular symmetry, then the family of Hamiltonians will also have the symmetry [10]. However, if the generator is replaced by something that is normal ordered w.r.t a particular expectation value, then the CUT RG flow will usually be towards either the symmetry-preserved state or the symmetry-broken state, depending on whichever is stable [11].

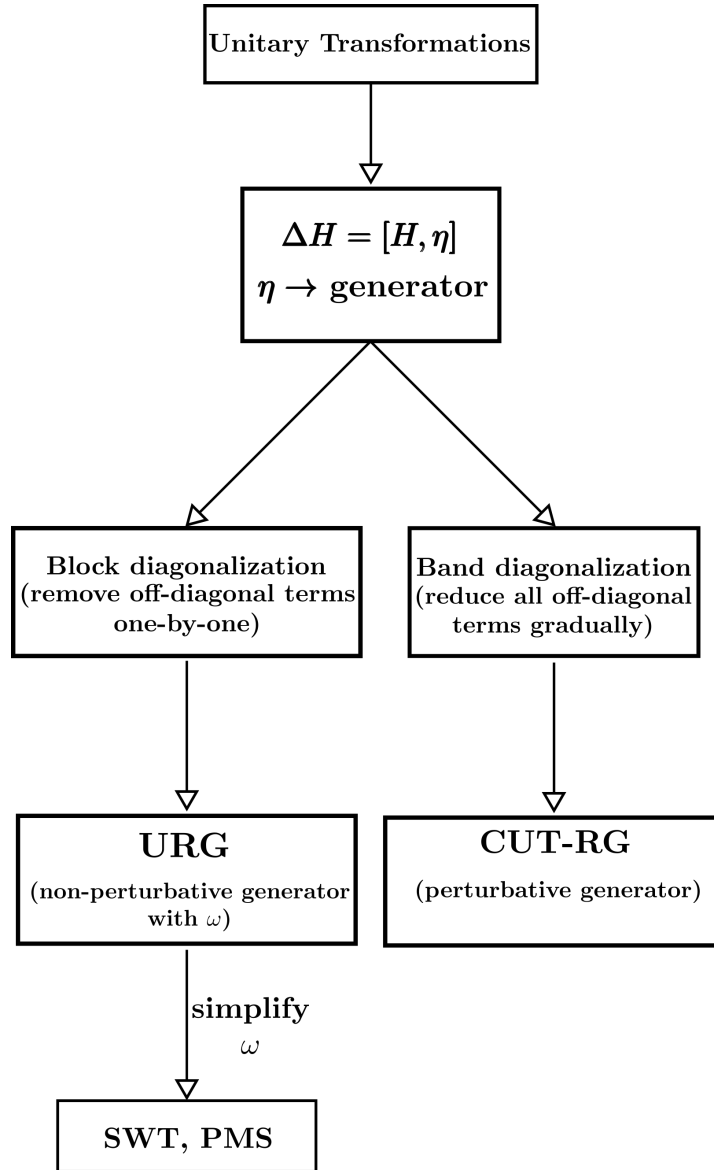


Figure 1.2: Comparison of the various unitary transformations and their relationships to each other.

1.5 Iterative Diagonalisation of Quantum Impurity Models

The broad idea is the following. We have a Hamiltonian describing an interacting set of states (in real or momentum space). We can express the total Hamiltonian in an incremental fashion: $H = \sum_i H_i$, where H_{i+1} involves more number of states than H_i . For example, a tight-binding model can be written in that form, with the definition $H_i = c_{i+1}^\dagger c_i + \text{h.c.}$. The iterative diagonalisation method obtains the low-energy spectrum of this problem in the following manner: We first diagonalise the Hamiltonian H for a smaller value of i , small enough such that this can be done exactly. We

then truncate the spectrum to a predefined size, and rotate all existing operators to this truncated basis, including the Hamiltonian H . We then consider the "bonding Hamiltonian" ΔH between the existing sites and the new sites, and rotate the same into the truncated basis. Adding the previous rotated Hamiltonian H and the rotated increment Hamiltonian ΔH gives us a truncated but effective Hamiltonian for the increased number of sites. We again diagonalise this, and again retain only a fixed number of states in the spectrum. We keep repeating this until we reach the required number of sites.

Jordan-Wigner Matrix Representation of Fermionic Operators

For a single qubit, the creation/annihilation matrices are

$$c = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, c^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (5.1)$$

given the convention that $|1\rangle = (0 \ 1)$ is the occupied state. For a many-body system, these must be replaced with field operators that have the canonical fermionic algebra. For a system of N 1-particle levels, this can be accomplished through a *Jordan-Wigner*-like transformation

$$c_j = \left(\otimes_1^j \sigma_z \right) \otimes c \otimes \left(\otimes_1^{N-j} \mathbb{I} \right), j \in [0, N-1], \quad (5.2)$$

where $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. For example, for spinless electrons on a three-site lattice, we can have three fermionic operators: c_1, c_2 , and c_3 . Following the above expression, their matrices are

$$c_1 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_3 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

For later use, we define the *antisymmetriser matrix* $S(j)$ and identity matrix $\mathbb{I}(j)$,

$$\begin{aligned} S(j) &= \otimes_1^j \sigma_z = \sigma_z \otimes \dots j \text{ times } \dots \otimes \sigma_z, \\ \mathbb{I}(j) &= \otimes_1^{N-j} \mathbb{I} = \mathbb{I} \otimes \dots j \text{ times } \dots \otimes \mathbb{I}, \end{aligned} \quad (5.3)$$

to express the fermionic representation compactly:

$$c_j = S(j) \otimes c \otimes \mathbb{I}(N-j). \quad (5.4)$$

Structure of Hamiltonian

We consider a general impurity model, of L number of sites (excluding the impurity site):

$$H_{\text{sys}}(L) = H_{\text{imp}} + H_{\text{imp-bath}} + -t \sum_{\sigma, j=1}^{L-1} \left(c_{j,\sigma}^\dagger c_{j+1,\sigma} + \text{h.c.} \right), \quad (5.5)$$

where H_{imp} is the Hamiltonian for the decoupled impurity site and $H_{\text{imp-bath}}$ is the impurity-bath coupling. Such a system has $2L + 2$ single-particle levels in the full system (2 spin levels for each of the $L + 1$ sites).

Iteration scheme (the initial number of states is L_0):

$$\begin{aligned} H_0(L_0) &= H_{\text{sys}}(L_0), \\ H_{r+1}(L_0) &= H_r(L_0) + \Delta H_r(L_0), \quad r \geq 0, \\ \Delta H_r(L_0) &= -t \sum_{\sigma} \left(c_{L_0+r,\sigma}^\dagger c_{L_0+r+1,\sigma} + \text{h.c.} \right). \end{aligned} \quad (5.6)$$

H_0 is the initial Hamiltonian, consisting of L_0 lattice sites in the bath, $H_{r+1}(L_0)$ is the Hamiltonian after $r + 1$ iterations having started with L_0 sites, and $\Delta H_r(L_0)$ is the increment term that gets added to $H_r(L_0)$ during the $(r + 1)^{\text{th}}$ iteration to give rise to $H_{r+1}(L_0)$.

Iterative Diagonalization

Let the starting Hamiltonian be $H_0(L_0)$, consisting of L_0 single-particle levels (hence a Hilbert space dimension of 2^{L_0}). We start with a single-particle computational basis and construct the L_0 fermionic operators $c_1, c_2, \dots, c_{L_0}$ in this basis, using eq. 5.4. We also keep track of a large antisymmetriser matrix $S(L_0)$ that will be used to attach new sites when we expand the system. Let M_s be the maximum number of eigenstates we retain in the spectrum at any given step. The value of M_s should be chosen so that a $M_s \times M_s$ matrix can be diagonalised in reasonable time.

- S1. Construct the complete Hamiltonian matrix H_0 in our present basis using the field operators $\{c_j\}$. Diagonalise the Hamiltonian (of size $D_0 \times D_0$) and obtain the eigenvalues E_n and eigenstates X_n . Each X_n is a column vector of size D_0 .
- S2. Retain at most M_s number of eigenstates, preferring the ones with lower energy. The reduced basis for this rotated truncated subspace is constructed by stacking the column vectors X_n horizontally:

$$R = [X_1 X_2 \dots X_{M_s}]_{D_0 \times M_s}. \quad (5.7)$$

This matrix also acts as the transformation to rotates and truncate all operators from the old basis into the new one.

- S3. We rotate our Hamiltonian H_0 , our fermionic operators $\{c_j\}$ and the large antisymmetrizer matrix $S(L_0)$ into the new reduced basis, using the transformation $O \rightarrow R^\dagger O R$.

- S4. We now need to expand our system by adding the increment Hamiltonian ΔH_0 . Let the number of *new* 1-particle levels in ΔH_0 be L_Δ . These new levels will be indexed as $L_0+1, \dots, L_0+L_\Delta$. We need to define antisymmetrized fermionic operators for the new sites:

$$\begin{aligned} c_{L_0+1} &= c \otimes \mathbb{I}(2^{L_\Delta-1}), \\ c_{L_0+2} &= \sigma_z \otimes c \otimes \mathbb{I}(2^{L_\Delta-2}), \\ &\dots \\ c_{L_0+L_\Delta} &= (\otimes_1^{L_\Delta-1} \sigma) \otimes c. \end{aligned} \tag{5.8}$$

These have to be calculated in the local computational basis (of size 2^{L_Δ}) of the new levels.

- S5. Combining the new sites with the old sites leads to a combined Hilbert space dimension of $(M_s + 2^{L_\Delta}) \times (M_s + 2^{L_\Delta})$. To allow all operators to act on the enlarged Hilbert space, we expand both sets of operators:

$$\begin{aligned} c_j &= S(L_0) \otimes c_j; \quad j = L_0 + 1, \dots, L_\Delta, \\ c_j &= c_j \otimes \mathbb{I}(2^{L_\Delta}); \quad j = 1, 2, \dots, L_0, \\ H_0 &\rightarrow H_0 \otimes \mathbb{I}(2^{L_\Delta}), \\ S(L_1) &= S(L_0) \otimes \mathbb{I}(2^{L_\Delta}). \end{aligned} \tag{5.9}$$

where $\mathbb{I}(2^{L_\Delta})$ is an identity matrix of dimension $2^{L_\Delta} \times 2^{L_\Delta}$. Note that the operators in the last three equations are the rotated ones (following Step 3).

- S6. Using the transformed operators $c_{L_0+1}, \dots, c_{L_0+L_\Delta}$ for the new sites, construct the difference Hamiltonian matrix ΔH_0 and hence the updated Hamiltonian $H_1 = H_0 + \Delta H_0$ for the next step. Repeat the process starting from step 2 with the new Hamiltonian H_1 , the new operators c_j and the new matrix $S(L_1)$ replacing the old counterparts.

1.5.1 Static correlations

We are in general interested in n -point correlations of the form $\langle O_1 O_2 \dots O_n \rangle$, where O_i are operators that act on 1-particle Hilbert spaces. Let the earliest step of the iterative diagonalisation procedure at which all these operators have entered the system be r . At this step r , construct the correlation operator $O_1 O_2 \dots O_n$ using the fermionic matrices c_j at that step (recall that these matrices will be highly rotated versions of the matrices we started with). Having constructed the operator O , we expand and rotate it after the completion of every future step: $O_{n+1} = (R_n^\dagger O_n R_n) \otimes \mathbb{I}(2^{L_\Delta})$, where R_n is the rotation matrix for the n^{th} step. The last step n^* of the iterative diagonalisation consists of only a diagonalisation and no expansion, resulting in a final set of eigenstates $\{X_i\}$. The form of the correlation operator in this basis is $O_{n^*} = R_{n^*-1}^\dagger O_{n^*-1} R_{n^*}$. The expectation value can now be calculated using the matrix O_{n^*} and the ground state of $\{X_i\}$.

1.5.2 Entanglement measures

In Section 1.7.1, we show how the matrix elements of reduced density matrices can be expressed as expectation values of certain local operators. Using such relations, the computation of reduced density matrix effective boils down to the calculation of specific correlations, which is covered in the previous section.

1.6 Optimising Fermi Surface Calculations

The below is an algorithm for reducing computational cost of carrying out iterative diagonalisation processes focused on the Fermi surface of a quantum impurity model, by rotating the interaction matrix to a sparser form. It is useful then the Hamiltonian has certain symmetry properties arising from specific lattice geometries.

Rotated basis for simplified representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the f -layer Fermi surface:

$$H^* = \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha,\beta} \sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} . \quad (6.1)$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4 . We assume the lattice structure is such that the following equation relates Q_1 with Q_3 and Q_2 with Q_4 :

$$J_{\mathbf{k},\mathbf{k}'} = -J_{\mathbf{k}+\mathbf{Q},\mathbf{k}'} = -J_{\mathbf{k},\mathbf{k}'+\mathbf{Q}}, \quad \mathbf{Q} = (\pi, \pi). \quad (6.2)$$

With this, we can write

$$\begin{aligned} \sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger + \sum_{\mathbf{k} \in Q_3, Q_4} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger + \sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k} - \mathbf{Q}_1, \mathbf{q}) f_{\mathbf{k}-\mathbf{Q}_1,\alpha}^\dagger + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k} + \mathbf{Q}_2, \mathbf{q}) f_{\mathbf{k}+\mathbf{Q}_2,\alpha}^\dagger \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k},\alpha}^\dagger - f_{\mathbf{k}-\mathbf{Q}_1,\alpha}^\dagger) + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k},\alpha}^\dagger - f_{\mathbf{k}+\mathbf{Q}_2,\alpha}^\dagger) \right] f_{\mathbf{q},\beta} . \end{aligned} \quad (6.3)$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k},\sigma}$:

$$A_{\mathbf{k},\sigma,\pm} = \begin{cases} \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}-\mathbf{Q}_1,\sigma}) , & \mathbf{k} \in Q_1 , \\ \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}+\mathbf{Q}_2,\sigma}) , & \mathbf{k} \in Q_2 , \end{cases} \quad (6.4)$$

which satisfy the appropriate algebra: $\{A_{\mathbf{k},\sigma,p}, A_{\mathbf{k}',\sigma',p'}\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'}\delta p,p'$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} = 2 \sum_{\mathbf{k},\mathbf{q} \in Q_1, Q_2} J_f^*(\mathbf{k},\mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} . \quad (6.5)$$

Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f_{\mathbf{k}\downarrow}^\dagger f_{\mathbf{k}'\uparrow} \rangle$), we use eq. 6.4 to express the bare fields $f_{\mathbf{k},\sigma}$ in terms of the new fields $A_{\mathbf{k},\sigma,\pm}$ and $B_{\mathbf{k},\sigma,\pm}$:

$$\begin{aligned} f_{\mathbf{k},\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} + A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_1, Q_2 \\ f_{\mathbf{k}-Q_1,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_1 \\ f_{\mathbf{k}+Q_2,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_2 \end{aligned} \quad (6.6)$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i,k_j} = \langle S_d^+ c_{k_i,\uparrow}^\dagger c_{k_j,\downarrow} \rangle$ for the bare fields f in terms of the rotated fields. Using the fact that the Hamiltonian does not couple the fields B_+ and A_+ , we have

$$\begin{aligned} F_{k_1,k_1} &= F_{k_3,k_3} = -F_{k_1,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \\ F_{k_2,k_2} &= F_{k_4,k_4} = -F_{k_2,k_4} = \frac{1}{2} \langle S_d^+ A_{k_2,\downarrow,-}^\dagger A_{k_2,\uparrow,-} \rangle , \\ F_{k_1,k_2} &= F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \end{aligned} \quad (6.7)$$

where the momentum state k_i belongs to the quadrant Q_i .

Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 6.5 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin (central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma_{i,\sigma}^\dagger = \sum_k \Lambda_{i,k} A_{k,\sigma}^\dagger , \text{ where } \sum_k \Lambda_{i,k}^* \Lambda_{j,k} = \delta_{i,j} . \quad (6.8)$$

These fermions of course satisfy the appropriate algebra:

$$\{\gamma_{i,\sigma}, \gamma_{j,\sigma'}^\dagger\} = \sum_{k,q} \Lambda_{i,k}^* \Lambda_{j,q} \{A_{k,\sigma}, A_{q,\sigma'}^\dagger\} = \delta_{i,j} \delta_{\sigma,\sigma'} . \quad (6.9)$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) A_{\mathbf{k}, \alpha}^\dagger A_{\mathbf{q}, \beta} = \sum_i K_i \gamma_{i, \alpha}^\dagger \gamma_{i, \beta} , \quad (6.10)$$

where K_i are the eigenvalues of the Kondo coupling matrix $J_f^*(\mathbf{k}, \mathbf{q})$. Substituting the definition of the γ fields (eq. 6.8) leads to the equation

$$J_f^*(\mathbf{k}, \mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda_{i,q}^\dagger \implies \mathcal{J} = \Lambda K \Lambda^\dagger , \quad (6.11)$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J_f^*(\mathbf{k}, \mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 6.11, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K . \quad (6.12)$$

The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha, \beta} \vec{\sigma}_{\alpha, \beta} \sum_i K_i \gamma_{i, \alpha}^\dagger \gamma_{i, \beta} . \quad (6.13)$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger . \quad (6.14)$$

1.7 Measures of entanglement and their computation

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [12–17]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [18–21]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [14, 22, 23].

1.7.1 Reduced density matrix elements as equal-time correlation functions

A density matrix plays the role of a probability density function (PDF) for a quantum system. A classical variable X that can take values $x \in [-X^*, X^*]$ has a PDF $F(X = x)$, which is used, for example, for calculating the average value (expectation value) of the variable:

$$\langle X \rangle = \int_{-X^*}^{X^*} x F(x) dx . \quad (7.1)$$

In quantum mechanics, the variables we are interested in are operators O , and a quantum PDF ρ will then give the expectation value of the operator, in a fashion similar to its classical counterpart:

$$\langle O \rangle = \sum_{\phi} \langle \phi | \rho O | \phi \rangle ; \quad (7.2)$$

here, the integral over the classical values x is replaced by a sum over the all possible quantum states $|\phi\rangle$ (complete orthonormal basis) of the system. However, we already know that the expectation value of a quantum operator is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle . \quad (7.3)$$

To bring this form closer to the one in terms of ρ (eq. 2), we insert an identity resolution: $1 = \sum_{\phi} |\phi\rangle \langle \phi|$:

$$\langle O \rangle = \sum_{\phi} \langle \Psi | O | \phi \rangle \langle \phi | \Psi \rangle = \sum_{\phi} \langle \phi | \Psi \rangle \langle \Psi | O | \phi \rangle . \quad (7.4)$$

A comparison of this equation to eq. (2) gives

$$\rho = |\Psi\rangle \langle \Psi| ; \quad (7.5)$$

a density matrix (also called density operator) is therefore formally defined as the outer product of the wavefunction with its conjugate. It encodes all information about the state; for example, the probability of measuring the system to be at a position x (hence in a state $|x\rangle$) is given by $P(x) = \langle x | \rho | x \rangle$. From its definition in eq. (5), the DM has the very important property that it has unity trace:

$$\text{Tr}[\rho] = 1 , \quad (7.6)$$

where $\text{Tr}[\cdot]$ is the trace of an operator. This is seen very easily by taking the trace in an orthonormal basis in which the state $|\Psi\rangle$ is one of the basis states. For example, consider the DM

$$\rho = \frac{1}{\sqrt{2}} [|\Psi_1\rangle \langle \Psi_1| + |\Psi_2\rangle \langle \Psi_2|] , \quad (7.7)$$

where $|\Psi_{1,2}\rangle$ are orthogonal normalised states. The trace of ρ^2 for such a state is only $1/2$; such states satisfying $\text{Tr}[\rho^2] < 1$ are termed *mixed states*; the DM describing such states cannot be written in the form $\rho = |\Psi\rangle \langle \Psi|$ for any wavefunction $|\Psi\rangle$. On the other hand, the DMs that *can* be expressed like that are termed *pure states*, and have $\text{Tr}[\rho^2] = 1$.

One way in which a pure state can be realistically converted into a mixed state is through a measurement process. Consider the zero total spin state formed by two spins:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle] , \rho = |\Psi\rangle \langle \Psi| . \quad (7.8)$$

The state $|\uparrow, \downarrow\rangle$ has the first spin pointing up and the second one down, and this is in a superposition with its flipped counterpart $|\downarrow, \uparrow\rangle$. This is clearly a pure state.

If we now measure the state of the second spin, it will collapse into one of its two states; the other spin will collapse into the opposite state (given the antiparallel configurations of the spins). What this means is that until one checks what the result of the measurement is, the first spin remains in a statistical superposition of both outcomes (up and down). This is formally expressed through a partial trace operation: the state of the first spin after a measurement of the second spin is obtained by taking the full density matrix ρ and computing its trace over the states $|\sigma_B\rangle$ of the second spin, leaving out the states of the first spin from the trace:

$$\rho_1 = \text{Tr}_2[\rho] = \sum_{\sigma_2} \langle \sigma_2 | \rho | \sigma_2 \rangle ; \quad (7.9)$$

where $|\sigma_2\rangle$ sums over the states $|\uparrow\rangle$ and $|\downarrow\rangle$ of the second spin. this partial tracing operation Tr_2 codifies the measurement process, and the partially traced density matrix ρ_1 is termed a reduced density matrix (RDM) of the first spin, because it only captures information about the first spin (the second spin has been traced out).

To see the connection with mixedness, let us compute the RDM for the first spin explicitly:

$$\rho_1 = \langle \uparrow_2 | \rho | \uparrow_2 \rangle + \langle \downarrow_2 | \rho | \downarrow_2 \rangle = \frac{1}{2} [|\uparrow_1\rangle \langle \uparrow_1| + |\downarrow_1\rangle \langle \downarrow_1|] , \quad (7.10)$$

where the numbered subscripts on the kets and bras indicate that they only act on the respective Hilbert spaces. To see that ρ_1 describes a mixed state, we adopt a specific representation:

$$\begin{aligned} |\uparrow_1\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} , |\downarrow_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} , \\ \Rightarrow \rho_1 &= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} , \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} . \end{aligned} \quad (7.11)$$

The trace of ρ_1^2 is half, hence it is a mixed state.

The deviation of trace of ρ^2 from its maximum value of unity can be thought of as a measure of the lost information. In the example above, when we traced out the states of the second spin, we lost some information about the state of the first spin as well. This is because the two spins were *entangled*; they had quantum correlations between them. The entanglement arises from the fact that the states of the two spins are anti-correlated.

If instead we had chosen a non-entangled parent state to start from, the reduced density matrix would not have been mixed. For example, consider the state

$$|\Psi\rangle = \frac{1}{2} (|\uparrow_1\rangle + |\downarrow_1\rangle) \otimes (|\uparrow_2\rangle + |\downarrow_2\rangle) . \quad (7.12)$$

It is easy to see that this state does not contain any correlations between the two spins - no matter the configuration of the second spin, the first spin is always in the state $|\uparrow_1\rangle + |\downarrow_1\rangle$. The RDM for the first spin comes out to be

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} , \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} , \quad (7.13)$$

revealing that ρ_1 is not mixed.

The above method of calculating an RDM works for simple cases but can fail if (i) we have fermion signs to account for, or (ii) the full basis cannot be expressed as a direct product of local basis sets. The former arises if we are dealing with fermions, and the latter arises if we are working in a rotated basis. We now derive an alternative method for calculating the RDM, one that automatically takes care of the fermion signs and works even if the basis is rotated and complicated to write down. We start by defining the RDM more precisely. Given a state $|\Psi\rangle$ and a bipartition into subsystems S_1 and its complement S_2 , the density matrix for the state is $\rho = |\Psi\rangle\langle\Psi|$. A basis state of the full system can be written as $|A, \alpha\rangle = |A\rangle \otimes |\alpha\rangle$, where $\{A\}$ and $\{\alpha\}$ are complete sets of basis states describing S_1 and S_2 respectively. The RDM for S_2 is then defined as

$$\rho_1 = \text{Tr}_\alpha \rho = \sum_\alpha \langle \alpha | \rho | \alpha \rangle . \quad (7.14)$$

The matrix elements of ρ_1 in the basis $\{A\}$ can be expressed as

$$\langle A | \rho_1 | B \rangle = \sum_\alpha \langle A, \alpha | \rho | B, \alpha \rangle . \quad (7.15)$$

We next define an operator $O_{A,B}$ that represents scattering processes between the various states of subsystem S_1 : $O_{A,B} = |A\rangle\langle B| \otimes I_2$, but acts as identity on S_2 . The expectation value of this operator is

$$\langle O_{A,B} \rangle = \text{Tr}(\rho O_{A,B}) = \sum_{C,\beta} \langle C, \beta | \rho | A \rangle \langle B | C, \beta \rangle = \sum_\beta \langle B, \beta | \rho | A, \beta \rangle . \quad (7.16)$$

Comparing with eq. 7.15, we obtain the relation

$$\langle A | \rho_1 | B \rangle = \langle O_{B,A} \rangle . \quad (7.17)$$

This result is significant because it guarantees that in order to calculate the RDM for a subsystem, it is enough to calculate the expectation values of all inter-state off-diagonal transition operators. For example, we take subsystem S_1 to be a two fermion system represented by the operators c_1 and c_2 . Assuming a basis $\{|0, 0\rangle, |0, 1\rangle, |1, 0\rangle, |1, 1\rangle\}$, where the two entries represent the occupancies of the two fermions, the RDM for this subsystem is determined by the following matrix:

$$\rho_1 = \begin{pmatrix} \langle h_1 h_2 \rangle & \langle h_1 c_2^\dagger \rangle & \langle c_1^\dagger h_2 \rangle & \langle c_1^\dagger c_2^\dagger \rangle \\ & \langle h_1 n_2 \rangle & \langle c_1^\dagger c_2 \rangle & \langle c_1^\dagger n_2 \rangle \\ & & \langle n_1 h_2 \rangle & \langle n_1 c_2^\dagger \rangle \\ & & & \langle n_1 n_2 \rangle \end{pmatrix} , \quad (7.18)$$

where the blank entries can be determined from hermiticity. To see how the operators are determined, take the example of the top right element. The two basis states for that matrix element are $|0, 0\rangle$ and $|1, 1\rangle$.

1.7.2 Quantum Fisher information: a witness of multipartite entanglement

The quantum Fisher information (QFI) F_Q [24, 25] quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle\langle\psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4 \left(\langle\psi|\hat{O}^2|\psi\rangle - \langle\psi|\hat{O}|\psi\rangle^2 \right), \quad (7.19)$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [24, 25]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. If $\varepsilon_m(\psi) \equiv 1 - |\langle m|\psi\rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4 \left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m \varepsilon_m \right)^2 \right]. \quad (7.20)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (7.20) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m \neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and a reduced susceptibility towards symmetry-breaking.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4 \left(\left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1^\dagger = \hat{O}_2} - \left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1 = 1} \right)^2 = 4 \left[G_{\hat{O}_2, \hat{O}_2}^<(t \rightarrow \infty) - \left(G_{1, \hat{O}_2}^<(t \rightarrow \infty) \right)^2 \right] \quad (7.21)$$

Further, the retarded Greens function defined above in eq. (??) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2}G_{O_2, O_2}^R(t \rightarrow \infty) = \langle (O_2)^2 \rangle$ and $\frac{i}{2}G_{1, O_2}^R(t \rightarrow \infty) = \langle O_2 \rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2iG_{O_2, O_2}^R(t \rightarrow \infty) + \left(G_{1, O_2}^R(t \rightarrow \infty)\right)^2. \quad (7.22)$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [25]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle \langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n} / \sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [25]

$$F_Q(\rho, \hat{O}) = 2 \sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\langle m | \hat{O} | n \rangle|^2. \quad (7.23)$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (7.24)$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ and the greater Greens function $G^>$ that we will now define admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G_{O^\dagger, O}^<(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G_{O^\dagger, O}^<(\omega) = 2\pi i \sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m | \hat{O}^\dagger | n \rangle|^2, \quad (7.25)$$

$$G_{O^\dagger, O}^>(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G_{O^\dagger, O}^>(\omega) = -2\pi i \sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (7.26)$$

If $\hat{O}$ is a Hermitian operator, eqs. (7.25) can be used to rewrite the QFI in terms of $G_{O^\dagger, O}^<(\omega)$. Indeed, by combining eqs. (7.25) and eq. (7.24) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \quad (7.27)$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \langle [\hat{O}(t), \hat{O}(t')] \rangle$. By going to the frequency domain, the imaginary part of such a function can

be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned}
 C_{\hat{O}}(\omega) &= i \int_0^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 \Rightarrow C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^\infty d(t-t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= i \int_{-\infty}^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= - \left(G_{\hat{O},\hat{O}}^>(\omega) + G_{\hat{O},\hat{O}}^<(\omega) \right) .
 \end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [25]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^\infty d\omega \tanh(\beta\omega/2) (C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*) . \quad (7.28)$$

1.7.3 Decomposition of entanglement entropy in terms of multipartite information I_n

We first define a partitioning of the entire system into N disjoint regions, $\mathcal{S}_N = \{\nu_1, \nu_2, \dots, \nu_N\}$. Going beyond bipartite mutual information, one can define the n -partite information I_n between the members of $\mathcal{S}_n = \{\nu_1, \nu_2, \dots, \nu_n\}$, which is itself a subset of the exhaustive set of regions $\mathcal{S}_N$:

$$I_n(\nu_1, \dots, \nu_n) = \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (7.29)$$

where σ is a subset drawn from the power set of $\mathcal{S}_n$, and $|\sigma|$ is the number of members in the subset σ . The 4-partite information can be written as

$$\begin{aligned}
 I_4(\nu_1, \nu_2, \nu_3, \nu_4) &= S_{EE}(\nu_1) + S_{EE}(\nu_2) + S_{EE}(\nu_3) + S_{EE}(\nu_4) - S_{EE}(\nu_1 \cup \nu_2) - S_{EE}(\nu_1 \cup \nu_3) \\
 &\quad - S_{EE}(\nu_1 \cup \nu_4) - S_{EE}(\nu_2 \cup \nu_3) - S_{EE}(\nu_2 \cup \nu_4) - S_{EE}(\nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3) \\
 &\quad + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_2 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3 \cup \nu_4) .
 \end{aligned} \quad (7.30)$$

One can also define the average n -partite information $\bar{I}_n$ by considering all sets of n regions:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma \in \mathcal{P}(S_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (7.31)$$

where $C_n(\mathcal{S}_N)$ represents all ways of choosing n distinct elements out of the full set $\mathcal{S}_N$ of subsystems.

The expression for $\bar{I}_n$ can be simplified by splitting the summation over the elements of the power set $\mathcal{P}$ into sets of definite number:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{m=1}^n \sum_{\sigma_m \in \mathcal{P}_m(S_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma_m) , \quad (7.32)$$

where $\mathcal{P}_m$ represents all subsets of the power set that have m elements. With some combinatorial magic, it can be shown that

$$\sum_{S_n \in C_n(S_N)} \sum_{\sigma_m \in \mathcal{P}_m(S_n)} = \binom{N-m}{n-m} \sum_{\sigma_m \in C_m(S_N)} . \quad (7.33)$$

To see how this comes about, note that what this equation says is that summing over combinations of subsets of length n of the super set S_N and further summing over all subsets of length $m \leq n$ for each set S_n is equivalent to summing just once over subsets of length m of the super set S_N but now each subset of length m appears with a frequency $f_m = \binom{N-m}{n-m}$. The value for the frequency f_m is simply the number of subsets S_n in which the smaller subset σ_m appears. This can be calculated by allowing m of the n slots in S_n to be filled by the members of σ_m , and the remaining $n - m$ slots then have to be filled by choosing from the remaining $N - m$ members of the super set S_N ; there's obviously $\binom{N-m}{n-m}$ ways of carrying this out.

Implementing this into the expression for $\bar{I}_n$ gives

$$\bar{I}_n = \sum_{m=1}^n (-1)^{1+m} \frac{\binom{n}{m}}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m) = \sum_{m=1}^n (-1)^{1+m} \binom{n}{m} S_{EE}^{(m)} , \quad (7.34)$$

where we have define the average entanglement entropy $S_{EE}^{(m)}$ for all m -member sets of regions:

$$S_{EE}^{(m)} = \frac{1}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m) . \quad (7.35)$$

With most of the machinery in place, we would now like to express the entanglement entropy S_{EE} purely in terms multipartite measures I_n , for the reasons stated above. For a system that is invariant under real-space translation (as our lattice model is), the entanglement entropy of any real-space interval should only depend on the extent of the interval. It therefore suffices to calculate the entanglement entropy averaged over all such intervals, and the corresponding measure in our formulation is $S_{EE}^{(1)}$.

We will now express $S_{EE}^{(1)}$ in terms of $\bar{I}_n$, by eliminating $S_{EE}^{(m)}$ ($m > 1$) from the equations. We adopt the ansatz

$$S_{EE}^{(1)} = \sum_{n=2}^N \lambda_n \bar{I}_n = \sum_{n=2}^N \lambda_n \binom{n}{m} \sum_{m=1}^n (-1)^{1+m} S_{EE}^{(m)} . \quad (7.36)$$

In order to separate $S_{EE}^{(1)}$ from the rest of the $S_{EE}^{(m)}$, we exchange the order of summation:

$$S_{EE}^{(1)} = \sum_{m=1}^N (-1)^{1+m} S_{EE}^{(m)} \sum_{n=m}^N \lambda_n \binom{n}{m} - S_{EE}^{(1)} \lambda_1 = S_{EE}^{(1)} \sum_{n=2}^N n \lambda_n - \sum_{m=2}^{N-1} S_{EE}^{(m)} (-1)^m \sum_{n=m}^N \binom{n}{m} \lambda_n , \quad (7.37)$$

where the constraint of $m \leq n$ in the previous form is now enforced by letting n run from m to N . The $N - 1$ coefficients $\{\lambda_n\}$ will be determined by requiring that the $S_{EE}^{(m)}$ ($m > 1$) vanish on the

right-hand side, as well as by matching the coefficients of $S_{\text{EE}}^{(1)}$ on both sides. Note that m runs only upto $N - 1$ in the second term; this is because, $S_{\text{EE}}^{(N)}$ is the entanglement entropy of the full system and hence automatically zero. The equations are

$$\sum_{n=2}^N n\lambda_n = 1, \quad (7.38)$$

$$\sum_{n=m}^N \binom{n}{m} \lambda_n = 0; \quad m = 2, \dots, N - 1. \quad (7.39)$$

We focus on the second equation first, since it is very short for m close to N . For $m = N - 1$ and $m = N - 2$, we get

$$\begin{aligned} m = N - 1 : \lambda_{N-1} + N\lambda_N &= 0 \implies \lambda_{N-1} = -N\lambda_N \\ m = N - 2 : \lambda_{N-2} + (N - 1)\lambda_{N-1} + \frac{N(N - 1)}{2}\lambda_N &= 0 \implies \lambda_{N-2} = \frac{N(N - 1)}{2}\lambda_N. \end{aligned} \quad (7.40)$$

We extrapolate this to the general solution,

$$\lambda_m = (-1)^m \binom{N}{m} \lambda_N, \quad m < N, \quad (7.41)$$

and then prove that if this is true for λ_{N-m} , it also solves the equation for λ_{N-m-1} , thus amounting to a proof by mathematical induction. The general equation for λ_{N-m-1} is

$$\sum_{n=N-m-1}^N \binom{n}{N-m-1} \lambda_n = 0 \implies \lambda_{N-m-1} + \sum_{m'=0}^m \binom{N-m'}{N-m-1} \lambda_{N-m'} = 0. \quad (7.42)$$

Substituting the extrapolated solution gives

$$\begin{aligned} \lambda_{N-m-1} &= - \sum_{m'=0}^m (-1)^{N-m'} \binom{N-m'}{N-m-1} \binom{N}{m'} \lambda_N = \binom{N}{m+1} \lambda_N \sum_{m'=0}^m (-1)^{N-m'-1} \binom{m+1}{m'} \\ &= (-1)^{N-m-1} \binom{N}{m+1} \lambda_N, \end{aligned} \quad (7.43)$$

which is consistent with the general solution eq. 7.41. This determines all coefficients λ_n for $n < N$ in terms of λ_N . To determine λ_N itself, we use eq. 7.38. Substituting eq. 7.41 into it gives

$$\lambda_N \sum_{n=2}^N n(-1)^n \binom{N}{n} = 1 \implies \lambda_N = \frac{(-1)^N}{N}. \quad (7.44)$$

The decomposition of entanglement entropy into multipartite measures is now complete:

$$\begin{aligned} S_{\text{EE}}^{(1)} &= \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N}{n} \bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(S_N)} I_n(S_n), \end{aligned} \quad (7.45)$$

where we have assumed N is even in order to drop the global phase factor $(-1)^N$.

1.8 Wavefunction Renormalisation Group

1.8.1 General idea

A reverse URG analysis involves studying the evolution of wavefunctions from the IR to the UV, by implementing inverse renormalisation group transformations. This is made possible by the fact that the IR fixed point Hamiltonians are often tractable and allow the computation of wavefunctions. Such an analysis gives insight into how states change as high-energy quantum fluctuations are resolved during the Hamiltonian renormalisation from the UV to the IR. It also allows studying the evolution of entanglement measures and correlation functions.

We will start with a very simple IR ground state wavefunction, and then go back towards the UV ground state by applying the inverse unitary operator $U^\dagger$:

$$\begin{aligned}
 U : \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}} &\rightarrow |1, 2, \dots, N-1\rangle |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} \\
 U^\dagger : \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} &\rightarrow |1, 2, \dots, N^*+1\rangle |N^*+2\rangle \dots |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}}
 \end{aligned}$$

The first process is the forward RG which we used to obtain the scaling equations. The second process is the reverse RG which we will undertake now. The method has already been applied to study the renormalisation of entanglement in various models of strong correlation. In general, we start with an IR wavefunction that consists a certain number of momentum states n_1 still entangled with the impurity, and the remaining momentum states n_2 disentangled from the impurity. The former are said to be part of the emergent Kondo cloud, while the latter are said to be part of the integrals of motion (IOMs) and appear in direct product with the cloud+impurity system in the ground state wavefunction. Each momentum state is tagged with two conduction bath levels, one above the Fermi surface and one below. These will be represented as $k, \pm$. If we represent the emergent cloud momentum states as $\{k_i\}$ and the IOM states as $\{q_i\}$, the ground state wavefunction can be written as

$$|\Psi\rangle_0 = \left(\bigotimes_{i=1}^{n_2} |q_i, -, \uparrow\rangle |q_i, -, \downarrow\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\bigotimes_{i=1}^{n_1} |q_i, +, \uparrow\rangle |q_i, +, \downarrow\rangle \right) \quad (8.1)$$

Throughout, we have assumed that the IOMs below the Fermi surface are occupied while those above are unoccupied. This means:

$$|\Psi\rangle_0 = \left(\bigotimes_{i=1}^{n_2} |\hat{n}_{q_i, -, \uparrow} = 1\rangle |\hat{n}_{q_i, -, \downarrow} = 1\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\bigotimes_{i=1}^{n_1} |\hat{n}_{q_i, +, \uparrow} = 0\rangle |\hat{n}_{q_i, +, \downarrow} = 0\rangle \right) \quad (8.2)$$

$|\Phi\rangle_{\text{cloud}}$ will be obtained by diagonalising a small system of n_1 conduction bath k states. This completes the construction of the IR ground state $|\Psi\rangle_0$. The algorithm of reverse RG then involves applying unitary transformations on this IR wavefunction; the unitary transformations will be the inverse of those that were applied during the forward RG algorithm. Since the forwards RG transformations decoupled k states and transformed them into the IOMS, the inverse transformations will

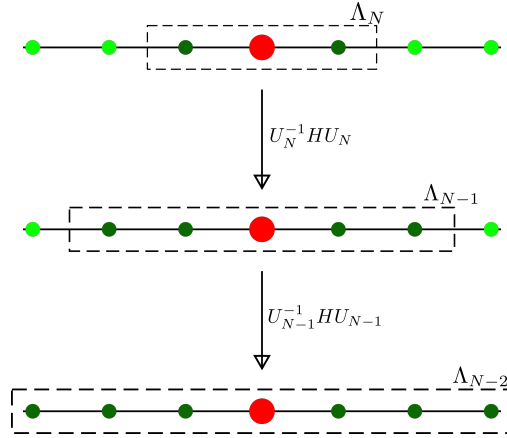


Figure 1.3: We start with a Hamiltonian with an impurity site (red) coupled with two conduction electrons (dark green), with four other decoupled electrons (bright green). The dotted rectangle represents the emergent window $(-\Lambda_j, \Lambda_j)$ at each step; the electrons inside that rectangle are still entangled with the impurity, while the ones inside have been decoupled. The next step of reverse RG involves applying the inverse transformation on the Hamiltonian, which will couple two more electrons from the IOMS (hence four dark green circles in the second step), leading to an enlargement of the emergent window. The unitary varies for each step, hence the notation U_j .

take the momentum states in the IOMs and re-entangle them with the impurity+cloud system, one at a time. This is shown in fig. 1.3.

The next step is to write down the unitaries that will take us from the IR ground state to the UV ground state. In the forward RG, we used the following unitary for decoupling the shell ϵ_q and spin β is

$$U_{q\beta} = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta} + \eta_{1\beta}^\dagger \right]. \quad (8.3)$$

Here, the subscripts 0 and 1 indicate it decouples an electron above and below the Fermi surface respectively. The inverse transformation for re-entangling $q\beta$ is

$$U_{q\beta}^\dagger = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta}^\dagger + \eta_{1\beta} \right]. \quad (8.4)$$

The wavefunction after reversing one step of the RG will thus be

$$|\Psi\rangle_1 = U_{q\uparrow}^\dagger U_{q\downarrow}^\dagger |\Psi\rangle_0 \quad (8.5)$$

Here, q is the first momentum state immediately outside the emergent cloud. Re-entangling further momentum states using their respective unitaries lead to the sequence of wavefunctions $|\Psi\rangle_2, |\Psi\rangle_3$ and so on. A straightforward way of calculating entanglement and correlations across the RG flow is to create a second-quantised prototypical form of the IR state $|\Psi_0\rangle$, and then apply the inverse unitary operators $\{U_i\}$ to this wavefunction in order to generate the family of states $|\Psi_i\rangle = U_i |\Psi_{i-1}\rangle, i > 0$. Quantities like entanglement entropy and correlations can then be calculated by using this family of states. If the unitary is of a specific form, there is instead a more efficient method of doing this, which is what we will describe in the next subsection.

1.8.2 Evolution equation for state tensor coefficients

In the following, we restrict ourselves to the reverse RG analysis of an impurity problem hybridising with the local degree of freedom of a conduction bath. At every step, we allow two states from the IOMS to interact with the already-entangled states (one above the Fermi energy and one below, in order to maintain particle-hole symmetry). After the j^{th} application of the reverse RG transformation, $2j$ number of states are entangled with the impurity site, and the state can be represented as

$$|\Psi_j\rangle = \sum_{\nu_d, \nu_1, \dots, \nu_{2j}} C_{\nu_d, \nu_1, \dots, \nu_{2j}, I_{2j+1}}^{(j)} |\nu_d\rangle |\nu_1\rangle |\nu_2\rangle \dots |\nu_{2j}\rangle \otimes |I_{2j+1}\rangle, \quad (8.6)$$

where $|\nu_d\rangle$ and $|\nu_i\rangle$ ($i \in [1, 2j]$) sum over the configurations of the impurity site and the effective zeroth site configuration at the j^{th} step. $|I_{2j+1}\rangle$ is the configuration of the integrals of motion from the $(2j+1)^{\text{th}}$ site and beyond. We define the product state $|\psi_{dz}(j)\rangle = |\nu_d\rangle |\nu_1\rangle \dots |\nu_{2j}\rangle$ for future reference.

We consider a restricted form of RG evolution, where the unitary transformation from IR towards UV can be expressed at any step as

$$U_j^\dagger = 1 + \sum_{i \in [1, 2j]} \sum_{\nu_d, \mu_d, \nu_i, \mu_i} \mathcal{T}_{\nu_d}^{\mu_d} (J_{\nu_d, \nu_i}^{\mu_d, \mu_i} c_{2j+1}^{\nu_i} \mathcal{T}_{\nu_i}^{\mu_i} + K_{\nu_d, \nu_i}^{\mu_d, \mu_i} c_{2j+2}^\dagger \mathcal{T}_{\nu_i}^{\mu_i}), \quad (8.7)$$

where $\mathcal{T}_\alpha^\nu = |\nu\rangle \langle \alpha|$ is a transition operator and sums over all transition operators of the specific subspace (impurity or bath sites). We first obtain the action of the transformation $U_j^\dagger$ on the basis states. We have

$$\begin{aligned} & \mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^\dagger \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \dots |\mu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle \\ &= \mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^\dagger (-1)^{\mu_d + \sum_{k < i} \nu_i} |\mu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle \\ &= \mathcal{T}_{\mu_d}^{\nu_d} (-1)^{\mu_d + \sum_{k < i} \nu_i} (-1)^{\mu_d + \sum_k \nu_i} |\mu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |1_{2j+2}\rangle \\ &= (-1)^{\sum_{k \geq i} \nu_i} |\nu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |1_{2j+2}\rangle, \end{aligned} \quad (8.8)$$

and

$$\mathcal{T}_{\mu_d}^{\nu_d} c_{2j+1} \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \dots |\mu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle = (-1)^{\sum_{k \geq i} \nu_i} |\nu_d\rangle \dots |\nu_i\rangle \dots |1_{2j+1}\rangle |0_{2j+2}\rangle. \quad (8.9)$$

Using these expressions, we construct the state at the $(j+1)^{\text{th}}$ step:

$$\begin{aligned} |\Psi_{j+1}\rangle &= \sum_{\nu_d, \nu_1, \dots, \nu_j} C_{\nu_d, \nu_1, \dots, \nu_j}^{(j)} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, \nu_d, \nu_i} C_{\mu_d, \dots, \mu_i, \dots}^{(j)} J_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\sum_{k \geq i} \nu_k} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, \nu_d, \nu_i} C_{\mu_d, \dots, \mu_i, \dots}^{(j)} K_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\sum_{k \geq i} \nu_k} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |1_{2j+2}\rangle |I_{2j+3}\rangle \end{aligned} \quad (8.10)$$

Gathering similar terms and normalising the state, we can write it as

$$|\Psi_{j+1}\rangle = \frac{1}{L_j} \sum_{v_d, \{v_i\}} |\psi_{dz}(j)\rangle \left[C_{v_d, \{v_i\}, 1_{2j+1}, 0_{2j+2}, I_{j+1}}^{(j)} |2_{2j+1}\rangle |0_{2j+2}\rangle + C_{+, v_d, \{v_i\}}^{(j)} |2_{2j+1}\rangle |1_{2j+2}\rangle \right. \\ \left. + C_{-, v_d, \{v_i\}}^{(j)} |1_{2j+1}\rangle |0_{2j+2}\rangle \right] |I_{2j+3}\rangle, \quad (8.11)$$

where $L_j^2 = \sum_{v_d, \{v_i\}} \left(C_{v_d, \{v_i\}}^{(j)}{}^2 + C_{+, v_d, \{v_i\}}^{(j)}{}^2 + C_{-, v_d, \{v_i\}}^{(j)}{}^2 \right)$ defines the normalisation factor, and

$$C_{+, v_d, \{v_i\}} = \sum_{i, \mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\sum_{k \geq i} v_k} \\ C_{-, v_d, \{v_i\}} = \sum_{i, \mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\sum_{k \geq i} v_k}. \quad (8.12)$$

We can now read off the renormalisation of the state tensor coefficients:

$$C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} = \frac{1}{L_j} C_{v_d, v_1, \dots, v_{2j}, 2_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j)}, \\ C_{v_d, v_1, \dots, v_{2j}, 2_{2j+1}, 1_{2j+2}, I_{2j+3}}^{(j+1)} = \frac{1}{L_j} \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i}, \quad (8.13) \\ C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} = \frac{1}{L_j} \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i},$$

We now make an important assumption: the coefficients $K_{\mu_d, \mu_i}^{v_d, v_i}$ and $J_{\mu_d, \mu_i}^{v_d, v_i}$ are uniform for the set of states $S(v_i)$ and zero for other states. That gives

$$C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 1_{2j+2}, I_{2j+3}}^{(j+1)} = \frac{1}{L_j} K \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\substack{\mu_d \in S(v_d), \\ \mu_i \in S(v_i)}} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)}, \quad (8.14) \\ C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} = \frac{1}{L_j} J \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\substack{\mu_d \in S(v_d), \\ \mu_i \in S(v_i)}} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)},$$

One can also look at renormalisation of expectation values of operators; this is useful for tracking the evolution of correlations and entanglement measures. Consider an operator $O_{\mu_k}^{v_k}$ that acts on the k^{th} bath site. The expectation value of this operator at any step can be written as

$$\langle O_{\mu_k}^{v_k} \rangle_j = \sum_{v_d, v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_{2j}} C_{v_d, \dots, \mu_k, \dots}^{(j)} C_{v_d, \dots, v_k, \dots}^{(j)}, \\ \langle O_{\mu_k}^{v_k} \rangle_{j+1} = \sum_{v_d, v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_{2j+2}} C_{v_d, \dots, \mu_k, \dots}^{(j+1)} C_{v_d, \dots, v_k, \dots}^{(j+1)} \quad (8.15)$$

1.9 Kondo model spectral function

1.9.1 Single-impurity Kondo model

For the Kondo model defined by the Hamiltonian

$$H_K = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \mathbf{S}_d \cdot \sigma_{\alpha,\beta} c_{k_1,\alpha}^\dagger c_{k_2,\beta}, \quad (9.1)$$

the impurity spin S_d does not have any fermionic operator. In order to define a spectral function for the impurity, we therefore look at the scattering of conduction electrons off it. This is captured by the conduction bath Greens function $G_{k,k'}$, which can be expressed in terms of a T -matrix when there are two-particle scattering processes present in the system:

$$G_{k,k'}(\omega) = G_k^{(0)}(\omega) + G_k^{(0)}(\omega) T_{k,k'}(\omega) G_{k'}^{(0)}(\omega), \quad (9.2)$$

where $T_{k,k'}$ is non-zero because of the presence of the impurity spin which scatters momentum states into each other. Because of the local nature of the impurity scattering, the T -matrix is also local: $T_{k,k'}(\omega) = T(\omega)$. We therefore use this to define a spectral function:

$$A_{\text{imp}} = -\frac{1}{\pi} \text{Im } T(\omega). \quad (9.3)$$

In order to calculate $T(\omega)$, we evaluate $G_{k,k'}$ using the equation of motion method. It is define as

$$G_{k,k'}(t) = -i\theta(t) \langle \langle c_{k,\sigma}(t), c_{k',\sigma}^\dagger \rangle \rangle. \quad (9.4)$$

Differentiating against time gives

$$\partial_t G_{k,k'}(t) = -i\delta(t) \langle \langle c_{k,\sigma}, c_{k',\sigma}^\dagger \rangle \rangle - \theta(t) \langle \langle [c_{k,\sigma}(t), H_K], c_{k',\sigma}^\dagger \rangle \rangle. \quad (9.5)$$

Evaluating the commutator gives

$$[c_{k,\sigma}(t), H_K] = \epsilon_k c_{k,\sigma} + J \sum_{q,\beta} \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{q\beta}. \quad (9.6)$$

For convenience we define the local bath operator $c_{0,\beta} = \sum_q c_{q,\beta}$. Substituting the commutator back into the expression gives

$$\partial_t G_{k,k'}(t) = -i\delta(t) \delta_{k,k'} - \theta(t) \epsilon_k \langle \langle c_{k,\sigma}(t), c_{k',\sigma}^\dagger \rangle \rangle - \theta(t) J \sum_{\beta} \langle \langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}, c_{k',\sigma}^\dagger \rangle \rangle(t). \quad (9.7)$$

Defining a new Greens function $\tilde{G}$,

$$\tilde{G}_{k'}(t) = -i\theta(t) \sum_{\beta} \langle \langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}, c_{k',\sigma}^\dagger \rangle \rangle, \quad (9.8)$$

we have the compact form

$$\partial_t G_{k,k'}(t) = -i\delta(t)\delta_{k,k'} - \theta(t)\epsilon_k \langle \langle c_{k,\sigma}(t), c_{k',\sigma}^\dagger \rangle \rangle - i\tilde{G}_{k'}(t). \quad (9.9)$$

We now transform to frequency space, using the definition $f(t) = \int d\omega e^{-i\omega t} g(\omega)$:

$$\omega G_{k,k'}(\omega) = \delta_{k,k'} + \epsilon_k G_{k,k'}(\omega) + J\tilde{G}_{k'}(\omega). \quad (9.10)$$

We setup the equation of motion for $\tilde{G}$ in a similar fashion. We use the Hermitian conjugate $-i\theta(t) \sum_\beta \langle \langle c_{k',\sigma}(t), \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}^\dagger \rangle \rangle$, because it is easier to work with:

$$\begin{aligned} \partial_t \tilde{G}_{k'}(t) &= -i\delta(t) \sum_\beta \delta_{\sigma,\beta} \langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta} \rangle - \theta(t) \langle \langle [c_{k',\sigma}(t), H_K], \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}^\dagger \rangle \rangle \\ &= -i\delta(t) \langle \mathbf{S}_d^z \rangle - \theta(t) \langle \langle (\epsilon_{k'} c_{k',\sigma}(t) + J \sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha} c_{0,\alpha}), \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}^\dagger \rangle \rangle \\ &= -i\delta(t) \langle \mathbf{S}_d^z \rangle - i\epsilon_{k'} G_{k'}(t) - J\theta(t) \langle \langle \sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha} c_{0,\alpha}, \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{0,\beta}^\dagger \rangle \rangle. \end{aligned} \quad (9.11)$$

At this point, we define a Kondo excitation operator $O_\sigma = J \sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha} c_{0,\alpha}$. Transforming to frequency space gives

$$\omega \tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \epsilon_{k'} G_{k'}(\omega) + \frac{1}{J} \langle \langle O_\sigma, O_\sigma^\dagger \rangle \rangle, \quad (9.12)$$

which gives an expression for $\tilde{G}_{k'}(\omega)$:

$$(\omega - \epsilon_{k'}) \tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \frac{1}{J} \langle \langle O_\sigma, O_\sigma^\dagger \rangle \rangle. \quad (9.13)$$

Substituting this into eq. 9.10 and noting that the non-interacting Greens function of the conduction bath is $G_k^{(0)} = (\omega - \epsilon_k)^{-1}$, we get

$$(G_k^{(0)})^{-1} G_{k,k'}(\omega) = \delta_{k,k'} + J \left[\langle S_d^z \rangle + \frac{1}{J} \langle \langle O_\sigma, O_\sigma^\dagger \rangle \rangle \right] G_{k'}^{(0)}, \quad (9.14)$$

which can be finally compared with the general expression in eq. 9.2:

$$G_{k,k'}(\omega) = G_k^{(0)} + G_k^{(0)} \left[J \langle S_d^z \rangle + \langle \langle O_\sigma, O_\sigma^\dagger \rangle \rangle \right] G_{k'}^{(0)}. \quad (9.15)$$

The T -matrix for scattering of conduction electrons off the impurity can be read off:

$$T_\sigma(\omega) = J \langle S_d^z \rangle + \langle \langle O_\sigma, O_\sigma^\dagger \rangle \rangle, \quad (9.16)$$

1.9.2 Two-impurity Kondo model

We now generalise this calculation to a two-impurity model S_1, S_2 , each with its own conduction bath:

$$H_{2K} = \sum_{k,\sigma,l} \epsilon_k n_{k,\sigma,l} + J_l \sum_{k_1,k_2,l,\sigma_1,\sigma_2} \mathbf{S}_l \cdot \sigma_{\sigma_1,\sigma_2} c_{k_1,\sigma_1,l}^\dagger c_{k_2,\sigma_2,l} + K \mathbf{S}_1 \cdot \mathbf{S}_2, \quad (9.17)$$

where l sums over the two spins and respective conduction baths, and K is the inter-impurity spin-exchange coupling.

Because of the composite nature of the impurity in this model, we consider a Greens function matrix

$$G = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}, \quad (9.18)$$

consisting of the diagonal Greens functions G_{ll} on the two impurities as well as an off-diagonal Greens function G_{12} . This is associated with a similar structure in the T -matrices:

$$T(k, k') = \begin{pmatrix} T(k, k')_{11} & T(k, k')_{12} \\ T(k, k')_{21} & T(k, k')_{22} \end{pmatrix}, \quad (9.19)$$

where the $T_{l,l'}$ are defined similar to the one-impurity case, but now with the additional index:

$$G_{(k,l),(k',l')}(\omega) = G_{k,l}^{(0)}(\omega) + G_{k,l}^{(0)}(\omega) T_{(k,l),(k',l')}(\omega) G_{k',l'}^{(0)}(\omega). \quad (9.20)$$

The additional physics here is the possibility of scattering from a momentum state k in the bath l to a momentum state k' in the other bath l' .

We can then use the T -matrices to define diagonal and off-diagonal spectral functions:

$$A_{\text{imp}}(l, l') = -\frac{1}{\pi} \text{Im } T_{l,l'}(\omega). \quad (9.21)$$

In order to calculate $T(\omega)$, we evaluate $G_{(k,l),(k',l')}$ using the equation of motion method. It is defined as

$$G(t) = -i\theta(t) \langle \langle c_{k,\sigma,l}(t), c_{k',\sigma,l'}^\dagger \rangle \rangle. \quad (9.22)$$

Differentiating against time gives

$$\partial_t G_{k,k'}(t) = -i\delta(t) \langle \langle c_{k,\sigma,l}, c_{k',\sigma,l'}^\dagger \rangle \rangle - \theta(t) \langle \langle [c_{k,\sigma,l}(t), H_{2K}], c_{k',\sigma,l'}^\dagger \rangle \rangle. \quad (9.23)$$

Evaluating the commutator gives

$$[c_{k,\sigma,l}(t), H_{2K}] = \epsilon_k c_{k,\sigma,l} + J_l \sum_{q,\beta} \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{q,\beta,l}. \quad (9.24)$$

For convenience we define the local bath operator $c_{0,\beta,l} = \sum_q c_{q,\beta,l}$. Substituting the commutator back into the expression gives

$$\begin{aligned} \partial_t G_{(k,l),(k',l')}(t) = & -i\delta(t) \delta_{k,k'} \delta_{l,l'} - \theta(t) \epsilon_k \langle \langle c_{k,\sigma,l}(t), c_{k',\sigma,l'}^\dagger \rangle \rangle \\ & - \theta(t) J_l \sum_{\beta} \langle \langle \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}, c_{k',\sigma,l'}^\dagger \rangle \rangle(t). \end{aligned} \quad (9.25)$$

Defining a new Greens function $\tilde{G}$,

$$\tilde{G}_{(0,l),(k',l')}(t) = -i\theta(t) \sum_{\beta} \langle \langle \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}, c_{k',\sigma,l'}^{\dagger} \rangle \rangle, \quad (9.26)$$

we have the compact form

$$\partial_t G_{(k,l),(k',l')}(t) = -i\delta(t)\delta_{k,k'}\delta_{l,l'} - \theta(t)\epsilon_k \langle \langle c_{k,\sigma,l}(t), c_{k',\sigma,l'}^{\dagger} \rangle \rangle - i\tilde{G}_{(0,l),(k',l')}(t). \quad (9.27)$$

We now transform to frequency space, using the definition $f(t) = \int d\omega e^{-i\omega t} g(\omega)$:

$$\omega G_{(k,l),(k',l')}(\omega) = \delta_{k,k'}\delta_{l,l'} + \epsilon_k G_{(k,l),(k',l')}(\omega) + J_l \tilde{G}_{(0,l),(k',l')}(\omega). \quad (9.28)$$

We setup the equation of motion for $\tilde{G}$ in a similar fashion. We use the Hermitian conjugate $-i\theta(t) \sum_{\beta} \langle \langle c_{k',\sigma,l'}(t), \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}^{\dagger} \rangle \rangle$, because it is easier to work with:

$$\begin{aligned} \partial_t \tilde{G}_{(0,l),(k',l')}(t) &= -i\delta(t)\delta_{l,l'} \sum_{\beta} \delta_{\sigma,\beta} \langle \mathbf{S}_l \cdot \sigma_{\sigma,\beta} \rangle - \theta(t) \langle \langle [c_{k',\sigma,l'}(t), H_{2K}], \sum_{\beta} \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}^{\dagger} \rangle \rangle \\ &= -i\delta(t)\delta_{l,l'} \langle \mathbf{S}_l^z \rangle - \theta(t) \langle \langle \epsilon_{k'} c_{k',\sigma,l'}(t) + J_{l'} \sum_{\alpha} \mathbf{S}_{l'} \cdot \sigma_{\sigma,\alpha} c_{0,\alpha,l'}, \sum_{\beta} \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}^{\dagger} \rangle \rangle \\ &= -i\delta(t) \langle \mathbf{S}_l^z \rangle - i\epsilon_{k'} \tilde{G}_{(0,l),(k',l')}(t) - J_{l'} \theta(t) \langle \langle \sum_{\alpha} \mathbf{S}_{l'} \cdot \sigma_{\sigma,\alpha} c_{0,\alpha,l'}, \sum_{\beta} \mathbf{S}_l \cdot \sigma_{\sigma,\beta} c_{0,\beta,l}^{\dagger} \rangle \rangle. \end{aligned} \quad (9.29)$$

At this point, we define a Kondo excitation operator $O_{\sigma,l} = J_l \sum_{\alpha} \mathbf{S}_l \cdot \sigma_{\sigma,\alpha} c_{0,\alpha,l}$. Transforming to frequency space gives

$$\omega \tilde{G}_{(0,l),(k',l')}(\omega) = \delta_{l,l'} \langle S_l^z \rangle + \epsilon_{k'} \tilde{G}_{(0,l),(k',l')}(\omega) + \frac{1}{J_l} \langle \langle O_{\sigma,l}, O_{\sigma,l}^{\dagger} \rangle \rangle, \quad (9.30)$$

which gives an expression for $\tilde{G}_{k'}(\omega)$:

$$(\omega - \epsilon_{k'}) \tilde{G}_{(0,l),(k',l')}(\omega) = \delta_{l,l'} \langle S_l^z \rangle + \frac{1}{J_l} \langle \langle O_{\sigma,l}, O_{\sigma,l}^{\dagger} \rangle \rangle. \quad (9.31)$$

Substituting this into eq. 9.10 and noting that the non-interacting Greens function of the conduction bath is $G_k^{(0)} = (\omega - \epsilon_k)^{-1}$, we get

$$(G_k^{(0)})^{-1} G_{(k,l),(k',l')}(\omega) = \delta_{k,k'}\delta_{l,l'} + J_l \left[\delta_{l,l'} \langle S_l^z \rangle + \frac{1}{J_l} \langle \langle O_{\sigma,l}, O_{\sigma,l}^{\dagger} \rangle \rangle \right] G_{k'}^{(0)}, \quad (9.32)$$

which can be finally compared with the general expression in eq. 9.2:

$$G_{(k,l),(k',l')}(\omega) = \delta_{k,k'}\delta_{l,l'} G_k^{(0)} + G_k^{(0)} \left[\delta_{l,l'} J_l \langle S_l^z \rangle + \langle \langle O_{\sigma,l}, O_{\sigma,l}^{\dagger} \rangle \rangle \right] G_{k'}^{(0)}. \quad (9.33)$$

The T -matrix for scattering of conduction electrons off the impurity can be read off:

$$T_{(\sigma,l),(\sigma,l')}(\omega) = \delta_{l,l'} J_l \langle S_l^z \rangle + \langle \langle O_{\sigma,l}, O_{\sigma,l}^{\dagger} \rangle \rangle, \quad (9.34)$$

1.10 Zero temperature Greens function in frequency domain

1.10.1 Spectral representation of Greens function

The impurity retarded Green's function is defined as

$$G_{d\sigma}(t) = -i\theta(t) \langle \{O_\sigma(t), O_\sigma^\dagger\} \rangle \quad (10.1)$$

where the average $\langle \rangle$ is over a canonical ensemble at temperature T , and O_σ is the excitation whose spectral function we are interested in. What follows is a standard calculation where we write the Green's function in the spectral representation. The ensemble average for the operator O_σ can be written in terms of the exact eigenstates of the Hamiltonian:

$$H|n\rangle = E_n|n\rangle, \quad \langle O_\sigma \rangle \equiv \frac{1}{Z} \sum_n \langle n|O_\sigma|n\rangle e^{-\beta E_n} \quad (10.2)$$

where $Z = \sum_n e^{-\beta E_n}$ is the partition function and $\{|n\rangle\}$ is the set of eigenfunctions of the Hamiltonian. We can therefore write

$$\begin{aligned} & \langle \{O_\sigma(t), O_\sigma^\dagger\} \rangle \\ &= \frac{1}{Z} \sum_m e^{-\beta E_m} \langle m| \{O_\sigma(t), O_\sigma^\dagger\} |m\rangle \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m| \left(O_\sigma(t) |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| O_\sigma(t) \right) |m\rangle \quad \left[\sum_n |n\rangle \langle n| = 1 \right] \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m| \left(e^{iH^*t} O_\sigma e^{-iH^*t} |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| e^{iH^*t} O_\sigma e^{-iH^*t} \right) |m\rangle \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \left(e^{i(E_m-E_n)t} \langle m| O_\sigma |n\rangle \langle n| O_\sigma^\dagger |m\rangle + e^{i(E_n-E_m)t} \langle m| O_\sigma^\dagger |n\rangle \langle n| O_\sigma |m\rangle \right) \\ &= \frac{1}{Z} \sum_{m,n} e^{i(E_m-E_n)t} ||\langle m| O_\sigma |n\rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \end{aligned} \quad (10.3)$$

The time-domain impurity Green's function can thus be written as (this is the so-called Lehmann-Kallen representation)

$$G_{d\sigma} = -i\theta(t) \frac{1}{Z} \sum_{m,n} e^{i(E_m-E_n)t} ||\langle m| O_\sigma |n\rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \quad (10.4)$$

We are interested in the frequency domain form.

$$\begin{aligned} G_{dd}^\sigma(\omega) &= \int_{-\infty}^{\infty} dt e^{i(\omega)t} G_{dd}^\sigma(t) \\ &= \frac{1}{Z} \sum_{m,n} ||\langle m| O_\sigma |n\rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega+E_m-E_n)t} \end{aligned} \quad (10.5)$$

To evaluate the time-integral, we will use the integral representation of the Heaviside function:

$$\theta(t) = -\frac{1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} \frac{1}{x + i\eta} e^{-ixt} dx \quad (10.6)$$

With this definition, the integral in $G_{d\sigma}(\omega)$ becomes

$$\begin{aligned} (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + i0^+ + E_m - E_n)t} &= (-i) \frac{-1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} \int_{-\infty}^{\infty} dt e^{i(\omega + i0^+ + E_m - E_n - x)t} \\ &= \frac{1}{2\pi} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} 2\pi \delta(\omega + i0^+ + E_m - E_n - x) \quad (10.7) \\ &= \frac{1}{\omega + E_m - E_n + i0^+} . \end{aligned}$$

The frequency-domain Green's function is thus

$$G_{dd}^{\sigma}(\omega) = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \frac{1}{\omega + E_m - E_n + i0^+} \quad (10.8)$$

The infinitesimally positive imaginary part in the denominator shifts the poles onto the lower half of the complex plane, leaving it analytic in the upper half: this is necessary to make the retarded Greens function causal.

1.10.2 Real and imaginary parts - The Sokhotski-Plemelj theorem

In order to write down the real and imaginary parts of the Greens function, we first prove the *Sokhotski-Plemelj* formula:

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x + i\eta} = P\left(\frac{1}{x}\right) - i\pi\delta(x) . \quad (10.9)$$

where $P(f(x))$ is the Cauchy principal value, defined as

$$P\left[\int_{-\infty}^{\infty} \frac{dx}{x}\right] = \int_{-\infty}^{0^-} \frac{dx}{x} + \int_{0^+}^{\infty} \frac{dx}{x} . \quad (10.10)$$

To prove the above identity, we integrate the left-hand side using a test function $f(x)$:

$$\begin{aligned} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)(x - i\eta)}{x^2 + \eta^2} \\ &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \left[\frac{x}{x^2 + \eta^2} - i\eta \frac{1}{x^2 + \eta^2} \right] \quad (10.11) \end{aligned}$$

The first integral can be split into three parts:

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \lim_{\eta \rightarrow 0^+} \lim_{\varepsilon \rightarrow 0^+} \left[\int_{-\infty}^{-\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{\varepsilon}^{\infty} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{-\varepsilon}^{\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} \right] \quad (10.12)$$

For the first two parts, it is safe to take the limit of η , because x is always non-vanishing there. For the third part, we can approximate $f(x)$ as $f(0)$ in the neighbourhood $x \in [-\varepsilon, \varepsilon]$, $\varepsilon \rightarrow 0$. This leaves an odd function $x/(x^2 + \eta^2)$ as the integrand, being integrated over a symmetric range. The third term therefore vanishes. In total, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \int_{-\infty}^{-\varepsilon} dx \frac{f(x)}{x} + \int_{\varepsilon}^{\infty} dx \frac{f(x)}{x} = P \left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x} \right). \quad (10.13)$$

To evaluate the second integral of eq. 10.11, we note that the only region in which the integrand is non-zero is when $|x| \simeq 0$; there, we again approximate $f(x)$ as $f(0)$. The remaining integral can then be evaluated easily:

$$-i \lim_{\eta \rightarrow 0^+} \eta \int_{-\infty}^{\infty} dx \frac{f(x)}{x^2 + \eta^2} = -i \lim_{\eta \rightarrow 0^+} \eta f(0) \int_{-\infty}^{\infty} \frac{dx}{x^2 + \eta^2} = -i\pi f(0), \quad (10.14)$$

where we used $\int \frac{dx}{x^2 + \eta^2} = \frac{1}{\eta} \arctan(x/\eta)$. Combining eqs. 10.11, 10.13 and 10.14, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} = P \left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x} \right) - i\pi \int_{-\infty}^{\infty} dx f(x) \delta(x). \quad (10.15)$$

This proves eq. 10.9.

The Sokhotski-Plemelj formula allows us to split the spectral representation into a real and an imaginary part:

$$\begin{aligned} G'_{d\sigma} &= \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) P \left(\frac{1}{\omega + E_m - E_n} \right) \\ G''_{d\sigma} &= -\pi \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n). \end{aligned} \quad (10.16)$$

The spectral function, specifically, is defined as follows:

$$A_{d\sigma}(\omega) = -\frac{1}{\pi} G'_{d\sigma} = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n) \quad (10.17)$$

1.10.3 Zero temperature spectral function

We specialise to zero temperature by taking the limit of $\beta \rightarrow \infty$. In both the partition function as well as inside the summation, the only term that will survive is the exponential of the ground state energy E_{GS} .

$$Z \equiv \sum_m e^{-\beta E_m} \implies \lim_{\beta \rightarrow \infty} Z = d_{GS} e^{-\beta E_{GS}}, \quad E_{GS} \equiv \min \{E_n\}$$

where d_{GS} is the degeneracy of the ground state. The spectral function then simplifies to

$$\begin{aligned} A_{d\sigma} &= \frac{1}{d_{GS} e^{-\beta E_{GS}}} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left[e^{-\beta E_m} \delta_{E_m, E_{GS}} + e^{-\beta E_n} \delta_{E_n, E_{GS}} \right] \delta(\omega + E_m - E_n) \\ &= \frac{1}{d_{GS}} \sum_{n, n_{GS}} \left[||\langle n_{GS} | O_{\sigma} | n \rangle||^2 \delta(\omega + E_{GS} - E_n) + ||\langle n | O_{\sigma} | n_{GS} \rangle||^2 \delta(\omega - E_{GS} + E_n) \right] \end{aligned} \quad (10.18)$$

The label n_{GS} sums over all states $|n_{\text{GS}}\rangle$ with energy E_{GS} . In practise, we evaluate this expression by replacing the formal Dirac delta functions with "nascent" ones, such as a Lorentzian function with a sufficiently sharp peak.

1.10.4 Reconstructing full Green function from spectral function: Kramers-Kronig relations

As mentioned earlier, the retarded Greens function $G_R(\omega + i0^+)$ has a complex pole in the lower half of the complex plane. In the following, the frequency ω is assumed to be complex. Consider the integral

$$I(\omega) = \oint_C d\omega' \frac{G_R(\omega')}{\omega' - \omega}, \quad \omega \in \mathbb{R}. \quad (10.19)$$

The contour C encloses the entirety of the upper half of the complex plane as well as almost all points of the real line but avoids the pole at $\omega' = \omega$ by forming a semicircle about that point that extends into the upper half of the plane. Since the integrand is completely analytic in the region enclosed by the contour (recall again that G_R is analytic in the upper half), the integral evaluates to zero (by Cauchy's theorem):

$$I(\omega) = 0. \quad (10.20)$$

One can also evaluate the integral along each part of the contour. The semicircular part contributes zero: this is because while the length of the semicircular arc goes as $|\omega'|$, the integrand vanishes faster than $1/|\omega'|$ as $|\omega'| \rightarrow \infty$ (this assumes that the Green function vanishes in the limit of $\omega \rightarrow \infty$). The part of the contour along the real axis can now be evaluated. The two parts of the contour on either side of the pole contribute

$$\int_{-\infty}^{\omega-0^+} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + \int_{\omega+0^+}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} \quad (10.21)$$

To evaluate the part of the integral along the small semicircular around $\omega' = \omega$ (that extends into the upper half), we argue that this contribution should be equal to that obtained by taking the semicircle that goes instead into the lower half but circulates in the opposite direction. Each of these contributions should, in turn, be equal to that obtained from joining these two contours. But joining these contours leads to a closed integral about the pole at $\omega' = \omega$, which is equal to $2\pi G_R(\omega)$.

$$\int_{\text{upper}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \int_{\text{lower}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \frac{1}{2} \oint_{C(\omega)} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = i\pi G_R(\omega). \quad (10.22)$$

Combining the last two equations, we get

$$I(\omega) = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + i\pi G_R(\omega). \quad (10.23)$$

Comparing with eq. 10.20 and separating into real part G'_R and imaginary part G''_R gives

$$P \int_{-\infty}^{\infty} d\omega' \frac{[G'_R(\omega') + iG''_R(\omega')]}{\omega' - \omega} + \pi [-G''_R(\omega) + iG'_R(\omega)] = 0. \quad (10.24)$$

Comparing the imaginary parts gives

$$G'_R(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{G''_R(\omega')}{\omega' - \omega} = -\mathcal{H} [G''_R(\omega)] , \quad (10.25)$$

where $\mathcal{H} [g(\omega)] = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{g(\omega')}{\omega - \omega'}$ is the Hilbert transform of the function $g(\omega)$. The spectral function was defined above as $-1/\pi$ times the imaginary part of the Greens function. Exchanging G''_R for A gives

$$G'_R(\omega) = \pi \mathcal{H} [A(\omega)] . \quad (10.26)$$

This provides a method for calculating the real part of a Greens function if the spectral function is already known.

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